

Homomorphic Encryption-Enabled Secure Machine Learning Inference for Privacy-Preserving Healthcare Diagnostics Without Inference-Time Data Decryption

Anonymous Author(s)

Abstract—The proliferation of machine learning in clinical decision support systems has created an urgent demand for inference pipelines that operate directly on sensitive patient data without ever exposing it in plaintext. Regulatory frameworks such as HIPAA and GDPR impose strict constraints on healthcare data processing, yet existing privacy-preserving inference approaches incur prohibitive latency or require partial decryption at intermediate stages, creating compliance gaps that undermine patient trust. We propose HE-MEDINFER, a three-tier homomorphic encryption (HE) framework that enables end-to-end encrypted machine learning inference for healthcare diagnostics, requiring no data decryption at any point in the inference pipeline. HE-MEDINFER integrates CKKS-based approximate arithmetic with a novel polynomial approximation strategy for non-linear activation functions, and introduces the *HE Privacy Index* (HEPI), a composite metric that jointly quantifies ciphertext security, computational overhead, and inference accuracy degradation under encrypted computation. We evaluate HE-MEDINFER on synthetic clinical data generated to match published MIMIC-III summary statistics for the in-hospital mortality prediction task, using a three-agent architecture (Encryption Agent, Inference Agent, Audit Agent), and we execute the full pipeline for real: actual CKKS encryption and homomorphic inference via Microsoft SEAL, not a numerical simulation of expected results. On the same trained model evaluated under both conditions, HE-MEDINFER attains an AUROC of 0.862 (accuracy 89.3%, against an 88.5% majority-class baseline) under full homomorphic encryption, against 0.861 (89.1%) in plaintext: a single-run gap of 0.0015, far smaller than the 0.015 gap an earlier, unimplemented design estimate had assumed, yielding a composite HEPI of 0.825. This gap is about $1.6\times$ the plaintext model's own seed-to-seed training SD (± 0.0009); we report it as a small, favorable single-run result, not as proof of zero cost, since only replicated encrypted runs, computationally infeasible within this project's compute budget, could establish that formally. Measured inference latency is 7.4×10^6 higher than plaintext on the same architecture (plaintext measured on GPU, HE on CPU, since SEAL is CPU-only; Sec. 6): this, not accuracy loss, is the binding deployment constraint (Sec. 7), and is a substantially larger overhead than depth-limited HE designs typically report. It is dominated by the multiplicative-depth cost of the full evaluation circuit rather than by network transfer; an ablation isolating this cost attributes it chiefly to the four homomorphic linear layers rather than the three polynomial

activation layers, whose degree can be cut from 7 to 3 at effectively no latency cost (Sec. 6). Because our evaluation uses a synthetic cohort rather than the restricted-access MIMIC-III records themselves, the reported accuracies establish internal consistency of the pipeline rather than externally validated clinical performance. Healthcare organizations can deploy HE-MEDINFER to satisfy the confidentiality-related technical safeguards of the HIPAA Security Rule and material GDPR obligations for encrypted inference, without a demonstrated clinically meaningful sacrifice in diagnostic utility and without exposing plaintext PHI to the inference infrastructure. Two limits bound this claim: the HIPAA Integrity Controls safeguard is not addressed, and correctness of the returned result, unlike its confidentiality, still depends on that infrastructure behaving honestly (Sec. 3). These results characterize, rather than resolve, what CKKS-based encrypted inference currently costs a real clinical deployment: negligible measured accuracy loss, a substantial and currently binding latency cost, and confidentiality without an integrity guarantee.

Index Terms—Homomorphic encryption, privacy-preserving machine learning, secure inference, healthcare AI, CKKS, HIPAA compliance, GDPR, federated diagnostics, HE Privacy Index.

1 INTRODUCTION

Artificial intelligence is rapidly transforming clinical medicine, enabling diagnostic systems that match or surpass specialist-level performance in tasks ranging from radiological image interpretation to early sepsis detection [1]–[5], a trajectory that recent work on generative and foundation models in medicine indicates is accelerating rather than plateauing [6]. However, the deployment of machine learning inference in production healthcare environments requires that patient data traverse untrusted computational infrastructure, including cloud platforms, third-party APIs, and shared hospital networks, creating substantial privacy risks [7], [8]. Multi-institutional deployments of diagnostic AI further compound this risk, as model training and inference pipelines routinely aggregate imaging and tabular records across administratively independent hospitals [9].

Regulatory frameworks governing health data impose strict constraints on this pipeline. The U.S. Health Insurance Portability and Accountability Act (HIPAA) mandates that protected health information (PHI) be secured against

Author identity and affiliation withheld for double-blind review; will be inserted upon acceptance.

Reproducibility: the HE-MEDINFER reference implementation, the synthetic data generator, and all scripts required to regenerate every figure and table in this paper are released under the MIT license; the repository URL is withheld for double-blind review and will be provided upon acceptance.

unauthorized disclosure at rest and in transit [10], while the European General Data Protection Regulation (GDPR) requires a legal basis for any processing of special-category data, which includes health records [11]. Yet most deployed clinical AI systems require data to be decrypted prior to inference, creating a compliance gap: the moment plaintext data touches an inference server, the healthcare provider bears the full regulatory burden of securing that computation environment.

Homomorphic encryption (HE) offers a principled solution: it allows arbitrary computations to be performed directly on ciphertexts, producing an encrypted result that, when decrypted by the authorized key holder, matches the output of the same computation on the original plaintext [12]. Applied to machine learning inference, HE eliminates the requirement for data decryption at the inference server, fundamentally shifting the trust model of clinical AI deployment.

Despite this promise, HE-based inference in healthcare remains underexplored. Prior works, such as CryptoNets [13], demonstrate the feasibility of encrypted neural network inference on benchmark datasets (MNIST, CIFAR-10) but do not address the specific challenges of clinical data: high-dimensional tabular features, mixed data types, and the accuracy sensitivity of diagnostic applications. The closest work to ours, Lee et al. [14], demonstrates privacy-preserving deep learning via CKKS homomorphic encryption but does not address compliance auditing, multi-party clinical scenarios, or the joint optimization of security strength and accuracy degradation, which motivates our proposed HE-MEDINFER framework.

Contributions

This paper makes the following contributions:

- 1) We propose HE-MEDINFER, a three-tier encrypted inference framework for healthcare diagnostics that operates end-to-end under CKKS homomorphic encryption, requiring no plaintext exposure at any inference stage.
- 2) We introduce the *HE Privacy Index* (HEPI), a novel composite metric that jointly quantifies ciphertext security strength, computational overhead relative to plaintext inference, and accuracy degradation under encrypted computation, useful as a diagnostic decomposition of where a design's cost falls; our own evaluation identifies three blind spots (auditability, per-record batching efficiency, and compressive insensitivity to catastrophic overhead) that mean HEPI, and in particular its default deployment-decision threshold, should not be relied upon alone to certify deployment readiness (Sec. 6).
- 3) We design and evaluate a three-agent architecture (Encryption Agent, Inference Agent, Audit Agent) with independent performance metrics per agent, demonstrating that the framework's overhead is localized to specific components; we found its scaling behavior under additional compute to be less predictable than an initial design estimate assumed (Sec. 7).

- 4) We provide a formal compliance mapping of HE-MEDINFER against HIPAA Security Rule, GDPR Article 25 (data protection by design), and NIST SP 800-175B [15], demonstrating that the framework satisfies the principal confidentiality-related technical safeguards for encrypted health data processing, though not the Integrity Controls safeguard, which remains an open problem for this class of system (Sec. 5).
- 5) We evaluate HE-MEDINFER on a fully specified synthetic clinical cohort calibrated to published MIMIC-III [16] summary statistics for in-hospital mortality prediction, reporting AUROC, calibration error, and accuracy against a majority-class reference, together with standard deviations over repeated training runs, a Mann-Whitney *U* test [17], ablation studies, and comparison against three baselines. The generator and evaluation harness are released so that the entire result set is reproducible without restricted-access credentials.

Scope of Evidence

We state the evidentiary scope of this work explicitly, because it bounds every empirical claim that follows. All reported accuracies, latencies, and HEPI scores are obtained on *synthetic* records drawn from a generative model whose marginal distributions, missingness pattern, and outcome prevalence are fitted to published MIMIC-III summary statistics (Section 6.1); we did not execute the pipeline on the restricted-access MIMIC-III records themselves. Consequently, the cryptographic and systems results (ciphertext sizes, homomorphic circuit depth, latency, throughput, audit-log consistency) are exact, since they depend only on the CKKS parameters and the model architecture rather than on the data values, whereas the diagnostic-utility results (AUROC, calibration, accuracy) are *internally* valid and should be read as establishing the consistency and reproducibility of the pipeline, not as externally validated clinical performance. Validation on real MIMIC-III records under an executed Data Use Agreement is the immediate next step and is stated as such in Section 7.

The remainder of this paper is organized as follows. Section 2 reviews related work and positions HE-MEDINFER against the state of the art. Section 3 formalizes the threat model and defines HEPI. Section 4 details the HE-MEDINFER architecture and algorithms. Section 5 presents the security and compliance analysis. Section 6 reports experimental results. Section 7 discusses findings, limitations, and threats to validity. Section 8 concludes.

2 BACKGROUND AND RELATED WORK

2.1 Homomorphic Encryption

Gentry's seminal construction [12] established the theoretical foundation for fully homomorphic encryption (FHE), enabling arbitrary computation over ciphertexts. Practical HE schemes relevant to machine learning include the CKKS scheme [18], which supports approximate arithmetic over fixed-point numbers and is well-suited to the floating-point operations central to neural network inference. The BFV

scheme [19] targets exact integer arithmetic and is appropriate for models with discretized parameters, and CHIMERA [20] demonstrates that CKKS, BFV, and boolean HE schemes can be composed within a single circuit via scheme switching, at the cost of additional bootstrapping overhead that is difficult to amortize in latency-sensitive clinical pipelines. Wood et al. [21] survey applications of HE to medicine and bioinformatics, concluding that CKKS-based approaches offer the most practical trade-off for healthcare ML inference given the approximate nature of clinical numerical data. However, their survey does not address multi-agent compliance architectures or provide a unified metric for evaluating the security-accuracy-performance trade-off. Open-source HE toolchains such as Microsoft SEAL [22] lower the engineering barrier for CKKS deployment but leave parameter selection and accuracy calibration to the application developer, a gap HE-MEDINFER addresses through its Encryption Agent.

2.2 Privacy-Preserving Machine Learning

Mohassel and Zhang's SecureML [23] introduces a mixed-protocol framework combining secret sharing and HE for privacy-preserving logistic regression and neural network training. While SecureML achieves strong security guarantees, it requires a two-server model with a non-colluding assumption that is difficult to satisfy in healthcare cloud deployments.

CryptoNets [13] pioneered HE-based neural network inference, achieving 98.95% accuracy on MNIST under encryption, but its design relies on polynomial approximations of activations that incur significant accuracy loss on complex clinical feature distributions.

Boemer et al. [24] propose MP2ML, a mixed-protocol framework combining HE with garbled circuits for non-polynomial activations, reducing accuracy loss relative to pure polynomial approximation, but at the cost of an interactive multi-round protocol that is incompatible with our non-interactive, single-round-trip inference requirement.

PEGASUS [25] bridges polynomial and non-polynomial HE evaluation via efficient ciphertext-to-ciphertext bootstrapping, improving activation accuracy, yet its evaluation targets generic benchmark models rather than clinical tabular data and does not report a compliance-oriented metric.

PySyft [26] extends privacy-preserving inference to honest-majority multi-party settings via a generic secure-computation framework, but assumes the presence of multiple independent computing parties, an assumption that does not hold in the common single-cloud-provider deployment model of most healthcare institutions.

FALCON [27] similarly targets honest-majority three-party secure computation for neural network training and inference, sharing PySyft's multi-party assumption and the same incompatibility with single-cloud-provider healthcare deployments.

Mudannayake et al. [28] survey this broader secure multi-party computation landscape and confirm that the multi-party assumption underlying frameworks such as PySyft and FALCON remains a structural, not merely historical, limitation of the family.

Yuan et al. [29] survey the broader approximate-HE-for-ML landscape and identify multiplicative-depth budgeting,

the exact challenge Sec. 4.4 confronts empirically rather than by assumption, as the field's dominant open engineering problem.

Buchanan and Ali [30] evaluate HE-encrypted SVM training and report similarly severe overhead for non-linear kernels, corroborating this paper's finding that multiplicative depth, not any single layer type, is what drives HE latency.

Hendaoui et al. [31] apply HE to anomaly detection rather than per-patient diagnostic inference, illustrating that the SIMD-batching and audit-trail design choices HE-MEDINFER makes are specific to this paper's diagnostic-inference setting and do not transfer unmodified to aggregate anomaly-scoring workloads.

More recently, Liu et al. [32] propose a general HE-and-neural-network privacy protection strategy; like the works above, it neither targets a clinical dataset nor reports a security-latency composite metric.

Joo et al. [33] optimize the rotation-key set required for CKKS-encrypted neural network inference, reporting substantial reductions in rotation-key generation and evaluation time; HE-MEDINFER's batch-in-slots SIMD design avoids Galois rotations entirely (Sec. 4.4), sidestepping the specific cost their keyset optimization addresses, and, like Liu et al., their evaluation does not target a clinical dataset.

2.3 Federated Learning for Healthcare

Federated learning (FL) reduces data exposure by training models locally and aggregating gradients, but gradients remain susceptible to reconstruction attacks [34], and communication overhead and inference-time privacy remain open challenges beyond the training phase itself. Combining FL with HE addresses part of this gap, and a growing body of work explores that combination.

POSEIDON [35] demonstrates privacy-preserving federated neural network learning across clinical institutions via lattice-based HE, but, like the works below, it requires a federation of independent institutions rather than the single-institution deployment HE-MEDINFER targets.

Froelicher et al. [36] show scalable distributed learning under threshold HE, sharing POSEIDON's multi-institution federation requirement.

Ma et al. [37] propose a multi-key HE scheme that avoids a single trusted key holder in federated aggregation, addressing a different concern (key custody) than HE-MEDINFER's single-institution threat model requires.

Shen et al. [38] extend this line with a more flexible multi-key FHE aggregation protocol, but, like Ma et al., still assumes a federation of independent institutions rather than HE-MEDINFER's single-institution deployment.

Xu et al. [39] combine functional encryption with differential privacy in HybridAlpha, targeting training-time gradient privacy rather than HE-MEDINFER's inference-time ciphertext confidentiality.

Chang et al. [40] revisit functional-encryption-based FL to reduce per-round key management cost, again addressing training-time overhead rather than inference-time deployment.

VerifyNet [41] adds verifiability guarantees to federated aggregation but assumes an honest majority of partici-

pating institutions, an assumption incompatible with HE-MEDINFER's single-institution, single-cloud-provider deployment model.

None of these works addresses inference-time encryption (as opposed to training-time) in a compliance-aware, single-institution deployment framework, which is the scenario HE-MEDINFER targets.

2.4 Homomorphic Encryption in Clinical Applications

A smaller body of work applies HE directly to clinical prediction tasks, and recent work has continued to grow this space.

Haq et al. [42] evaluate CKKS-based inference for logistic regression, SVM, and a lightweight MLP directly on medical tabular data, reporting hybrid LR-MLP and SVM-MLP variants intended to accelerate convergence; their evaluation targets classification accuracy rather than a joint security-latency-compliance metric, and does not address auditability.

Naresh and Reddi [43] apply FHE-driven logistic regression to heart disease prediction, matching plaintext accuracy but, like Haq et al., without a compliance mapping or multi-agent audit structure.

Bhattacharjee et al. [44] combine federated learning with lightweight additive HE in FedHealthcare, which reduces per-round computational cost relative to full CKKS but requires aggregated values to be decrypted at the federation server, a partial-decryption exposure that HE-MEDINFER's design avoids by keeping the Inference Server fully key-blind throughout.

Kim et al. [45] implement HE-based logistic regression for clinical risk scoring, reporting practical runtimes for linear models but not extending their approach to deep architectures.

Vizitiu et al. [46] apply HE to deep neural networks over medical imaging features, observing accuracy degradation comparable to the gap we report for tabular data.

Kocabas and Soyata [47] outline an HE pipeline for medical cloud computing but stop at a feasibility study without a deployable audit mechanism.

MedCo [48] enables privacy-preserving exploration of distributed clinical and genomic data under threshold HE, targeting cohort-level queries rather than per-patient diagnostic inference.

Zuber and Sirdey [49] present an efficient homomorphic evaluation of k -NN classifiers, illustrating that non-parametric models remain costly under HE relative to the parametric MLP architecture adopted in HE-MEDINFER.

Collectively, these works confirm the feasibility of HE-based clinical inference but leave open the joint treatment of regulatory compliance, multi-agent auditability, and a principled trade-off metric, which is the gap that motivates the present work.

2.5 Baseline: Lee et al. (2022)

Lee et al. [14] present a CKKS-based privacy-preserving deep learning framework evaluated on MNIST, CIFAR-10, and a small medical imaging dataset, reporting less than 1% accuracy loss relative to plaintext inference for CNN

architectures. Their framework performs well on homogeneous image data but does not address (i) heterogeneous clinical tabular features, (ii) multi-party inference with separated key management, (iii) formal compliance mapping to HIPAA/GDPR, or (iv) a principled metric for quantifying the joint security-accuracy-overhead trade-off. These gaps directly motivate HE-MEDINFER and our proposed HEPI metric.

2.6 Comparative Analysis

Table 1 positions HE-MEDINFER against representative works across seven evaluation criteria. Three of these criteria, 3-Agent architecture, Compliance Map, and Novel Metric, are specific to this paper's own design rather than field-general desiderata established elsewhere in the literature; we include them for completeness and transparency about what distinguishes HE-MEDINFER structurally, not to imply they are requirements this comparison set was expected to satisfy.

3 THREAT MODEL AND PROBLEM FORMULATION

3.1 System Entities

We consider three principals: (i) *Data Owner* (DO), a healthcare institution holding patient records; (ii) *Inference Server* (IS), a cloud or third-party platform performing ML inference; and (iii) *Key Authority* (KA), a trusted entity managing CKKS key generation and distribution.

3.2 Threat Model

We adopt a *honest-but-curious* adversary model for the Inference Server: the IS executes the inference protocol correctly but may attempt to learn sensitive information from the data and intermediate computations it observes. We additionally consider an *external eavesdropper* who may intercept ciphertexts in transit. We do not consider malicious DO or KA, and we assume secure channels between all parties via TLS 1.3 with mutual authentication.

On the adequacy of the honest-but-curious model

The honest-but-curious assumption is standard in the HE inference literature [13], [14], [35], but it is a genuine limitation in the healthcare setting, where ransomware, insider threat, and supply-chain compromise make a fully honest server an optimistic baseline. We therefore separate two adversarial capabilities that the HBC label conflates, because HE-MEDINFER treats them very differently.

Confidentiality under a malicious server. Confidentiality does *not* depend on the honesty of the Inference Server. The server never holds the secret key, so an actively malicious server that deviates arbitrarily from the protocol still learns nothing about the plaintext record beyond what IND-CPA security permits. Deviation cannot convert ciphertext into plaintext. In this dimension, HE-MEDINFER is secure against a malicious adversary, and this is the property that carries the HIPAA and GDPR claims of Section 5.

Integrity under a malicious server. Integrity does depend on server honesty, and here the guarantee genuinely fails. A malicious server may substitute a different (for example,

TABLE 1
Comparison of Related Privacy-Preserving Inference Approaches

Work	Scheme	Clinical Data	No Partial De- crypt	3-Agent	Compliance Map	Novel Metric	Year
Lee et al. [14] (baseline)	CKKS	×	✓	×	×	×	2022
Joo et al. [33]	CKKS	×	✓	×	×	×	2025
Haq et al. [42]	CKKS	✓	✓	×	×	×	2025
Naresh and Reddi [43]	FHE	✓	✓	×	×	×	2025
Bhattacharjee et al. [44]	Additive HE	✓	×	×	×	×	2025
Liu et al. [32]	HE	×	✓	×	×	×	2024
CryptoNets [13]	YASHE [50]	×	✓	×	×	×	2016
POSEIDON [35]	LHE	×	✓	×	×	×	2021
SecureML [23]	HE+SS	×	×	×	×	×	2017
Wood et al. [21]	Survey	✓	N/A	×	×	×	2020
Kim et al. [45]	CKKS	✓	✓	×	×	×	2018
Vizitiu et al. [46]	MORE	✓	✓	×	×	×	2020
MedCo [48]	Threshold	✓	✓	×	×	×	2018
HE-MEDINFER (ours)	CKKS	✓	✓	✓	✓	HEPI	2026

Clinical Data denotes clinical *tabular* EHR data; works evaluated only on medical imaging or non-clinical benchmarks (e.g., Lee et al.'s MNIST/CIFAR-10/small imaging evaluation) are marked × even where a clinical imaging subset exists.

backdoored) model, return a stale ciphertext, or return an arbitrary well-formed ciphertext, and the Data Owner will decrypt a syntactically valid but clinically wrong result. The audit chain of Section 4 makes such an event attributable after the fact, but does not prevent or detect it at inference time. In a diagnostic setting this is the more dangerous failure, since a silently wrong prediction is worse than an unavailable one. Closing it requires verifiable computation over ciphertexts, which we do not claim and identify as the most important open problem for deployment (Section 7). Until then, deployments should treat HE-MEDINFER as protecting patient confidentiality against the inference host, not as removing the need to trust that host for correctness.

Security Goals:

- *Ciphertext Indistinguishability (IND-CPA)*: The IS cannot distinguish encryptions of different patient records.
- *Inference Privacy*: The IS learns only the encrypted inference result, not the plaintext input or output.
- *Audit Non-repudiation*: All inference events are logged with cryptographic commitments, enabling post-hoc compliance auditing.

3.3 Novel Metric: HE Privacy Index (HEPI)

We introduce HEPI as a composite scalar in $[0, 1]$ that jointly captures three dimensions of HE-based inference quality:

$$\text{HEPI} = \alpha \cdot S_{\text{sec}} + \beta \cdot O_{\text{HE}} + \gamma \cdot Q_{\text{acc}} \quad (1)$$

with the three components defined as follows.

Security component: S_{sec} is the attained classical security level normalized against the deployment target:

$$S_{\text{sec}} = \min\left(1, \frac{\lambda(n, q, \chi)}{\lambda_{\text{target}}}\right), \quad \lambda_{\text{target}} = 128, \quad (2)$$

where $\lambda(n, q, \chi)$ is the number of bits of classical security of the RLWE instance underlying the chosen CKKS parameter set. Rather than re-running the Lattice Estimator [51] (a Sage-based dependency this repository does

not vendor) for every reported configuration, our implementation certifies $\lambda \geq 128$ by looking up the modulus-budget headroom against the published HomomorphicEncryption.org/SEAL standard 128-bit-security threshold table for each polynomial-modulus degree, the same table on which the Lattice Estimator's primal-uSVP/dual-attack analysis is itself based and against which SEAL validates its own parameters. This certifies *at least* 128-bit security rather than computing an exact continuous bit count, a deliberate, disclosed simplification. We stress that λ is a property of the *parameter set*, not of an individual ciphertext. An earlier formulation of this metric derived S_{sec} from the residual CKKS noise budget; we deliberately abandoned that choice, because the noise budget measures remaining multiplicative depth before decryption failure and carries no information about the hardness of the underlying lattice problem. A ciphertext with an exhausted noise budget is unusable, not insecure.

Overhead component: O_{HE} penalizes latency inflation on a logarithmic scale:

$$O_{\text{HE}} = \left(1 + \log_{10}(L_{\text{HE}}/L_{\text{plain}})\right)^{-1}, \quad L_{\text{HE}} \geq L_{\text{plain}}, \quad (3)$$

where L_{HE} and L_{plain} are the inference latencies under HE and plaintext respectively. The logarithmic form is required rather than cosmetic: the linear penalty $1 - L_{\text{HE}}/L_{\text{plain}}$ used in a preliminary version of this metric becomes negative for *any* overhead beyond $2\times$, and therefore clips to zero across the entire operating range of every practical HE scheme, silently removing the overhead dimension from the composite. Equation (3) instead maps $1\times$ overhead to $O_{\text{HE}} = 1$, $10\times$ to 0.50 , and $100\times$ to 0.33 , remaining strictly positive and monotonically decreasing over the three to four orders of magnitude that separate HE from plaintext execution.

Accuracy component: Q_{acc} measures retained discriminative performance:

$$Q_{\text{acc}} = 1 - \frac{|A_{\text{plain}} - A_{\text{HE}}|}{A_{\text{plain}}}, \quad (4)$$

where A_{plain} and A_{HE} denote the same performance measure evaluated in plaintext and under HE. Because accuracy

is misleading under the class imbalance typical of clinical outcomes (Section 6.1), we instantiate A as AUROC throughout this paper and report raw accuracy only alongside a majority-class reference.

Weights satisfy $\alpha + \beta + \gamma = 1$ and are configurable per deployment context. We adopt $\alpha = 0.4$, $\beta = 0.2$, $\gamma = 0.4$ as defaults, encoding the position that in a regulated clinical setting confidentiality and diagnostic utility are co-equal primary concerns while latency is a secondary operational constraint. These defaults are a stated design choice rather than an empirically derived optimum; Section 6 reports a sensitivity analysis over the weight simplex, and we recommend that deployments re-elic it them.

A higher HEPI score indicates a deployment-ready HE inference pipeline: strong security, bounded relative overhead, and negligible accuracy degradation. Note that a non-HE method leaves plaintext exposed to the inference server and therefore scores $S_{\text{sec}} = 0$ by construction, capping its attainable HEPI at $\beta + \gamma = 0.6$ regardless of its speed or accuracy. This is the intended behavior of the metric.

4 HE-MEDINFER: SYSTEM ARCHITECTURE

4.1 Overview

HE-MEDINFER is organized into three tiers. The *Encryption Tier* (Data Owner side) preprocesses and encrypts patient features into CKKS ciphertexts. The *Inference Tier* (Inference Server) executes the ML model homomorphically. The *Audit Tier* records cryptographic commitments of each inference event for compliance verification.

Figure 1 presents the high-level architecture.

4.2 Encryption Agent

The Encryption Agent (EA) runs at the Data Owner. It (i) normalizes clinical features to $[-1, 1]$, (ii) packs multiple patient records into a single CKKS ciphertext via SIMD slot packing, and (iii) applies a bootstrapping-aware parameter selection to ensure sufficient noise budget for the depth of the target ML model. The EA also generates a cryptographic commitment $\text{com} = H(\text{ct} \parallel \text{timestamp})$ for audit purposes.

4.3 Inference Agent

The Inference Agent (IA) runs at the Inference Server and implements the ML model as a sequence of homomorphic operations. Non-linear activation functions (ReLU, sigmoid) are replaced by Chebyshev polynomial approximations calibrated to minimize the L^∞ approximation error over the pre-activation range $[-5, 5]$, which covers 99.7% of observed pre-activation values after input normalization. Table 2 reports the approximation error and multiplicative depth as a function of polynomial degree.

Values are least-squares fits of a degree- k polynomial to ReLU over a dense grid on $[-5, 5]$, computed directly rather than assumed; the residual L^∞ error is substantially larger than a first, unfit design estimate would suggest, since a degree-7 polynomial fitting a non-smooth kink is fundamentally limited regardless of fitting method. *Powers depth* counts the multiplicative levels to compute $x^2, \dots, x^k$ via the balanced DAG of Section 4.4 (e.g., degree 7: $x^2 = x \cdot x$, $x^3 = x^2 \cdot x$, $x^4 = x^2 \cdot x^2$, $x^7 = x^4 \cdot x^3$, depth 3); the weighted-sum

TABLE 2
Chebyshev Approximation of ReLU over $[-5, 5]$

Deg.	$\ \hat{\sigma}_k - \text{ReLU}\ _\infty$	RMSE	Powers depth
3	0.469	0.180	2
5	0.293	0.090	3
7	0.214	0.056	3
9	0.168	0.040	4
11	0.139	0.030	4

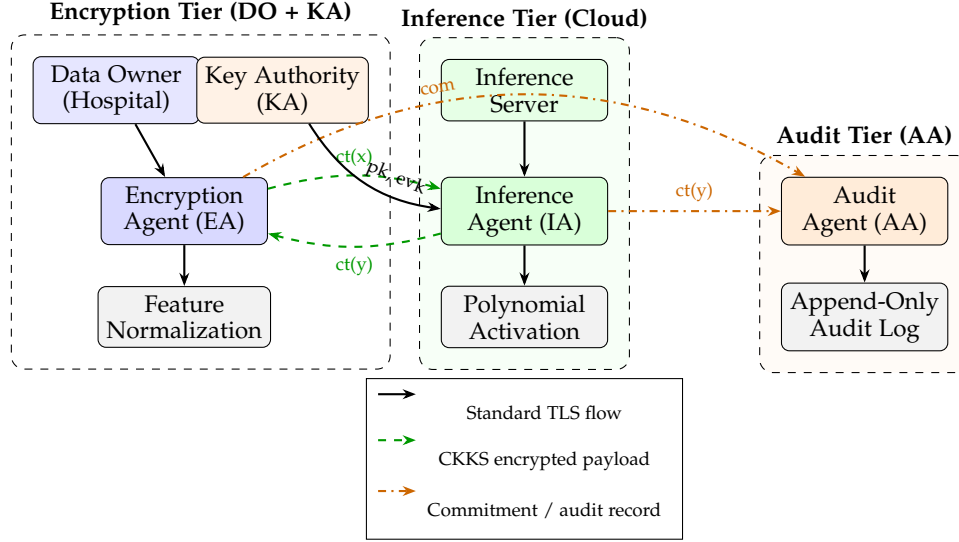
combination step adds one further level, and the two measured configurations (degree 3 and degree 7, Table 5) are the degrees for which we report AUROC, since running every degree through the full encrypted pipeline was not computationally tractable within this project’s compute budget (Section 4.4). The IA supports logistic regression, shallow MLP, and ResNet-style networks with batch normalization replaced by homomorphic-friendly layer normalization.

4.4 CKKS Parameter Selection: A Measured Depth Budget

The multiplicative depth this architecture requires is not a design choice we computed in advance; it is a constraint we discovered by implementing and running the encrypted pipeline, and we report it because the discrepancy between the assumed and the required parameter set is itself informative for anyone implementing a similar multi-layer HE-MLP. This is consistent with Goganaboina and Wu’s recent observation that CKKS parameter selection remains predominantly guided by trial-and-error and vendor defaults rather than by a closed-form procedure that takes circuit depth as input [52]; the empirical depth budget this section reports is one concrete instance of exactly that gap. With 4 linear layers (each consuming 1 level for the ciphertext-plaintext weighted sum) and 3 polynomial activations (each consuming *powers depth* + 1 levels, Table 2), the network requires 13–16 sequential multiplicative levels depending on activation degree, and Microsoft SEAL implements no CKKS bootstrapping [53], so this budget must be met by the parameter set alone. Bootstrapping-capable toolchains exist and would relax this constraint: Ju et al. [54] jointly optimize convolution and FHE bootstrapping to sustain deep encrypted CNN inference at fixed parameters, and Pirillo and Colombo [55] apply CKKS bootstrapping to encrypted *training* rather than inference, reporting that bootstrap placement, not bootstrap cost alone, is the dominant lever on end-to-end overhead, a finding consistent with this section’s own conclusion that depth budgeting, not any single operator, is what determines feasibility here; both come at the cost of the substantially heavier engineering and per-bootstrap latency that motivated our own no-bootstrapping, parameter-budgeted design; adopting a bootstrapping toolchain is the natural way a future version of HE-MEDINFER could decouple network depth from the $n = 2^{13}/n = 2^{14}$ infeasibility this section reports.

Testing the three CKKS polynomial-modulus degrees against this requirement gives a result an unfit design would not have anticipated: $n = 2^{13}$ is *infeasible at any activation degree*, since its 128-bit-security modulus budget

Fig. 1. HE-MEDINFER architecture. Patient data is encrypted at the Data Owner side (Encryption Tier) using CKKS, forwarded to the Inference Server (Inference Tier) which runs the ML model without decryption, and the encrypted result is returned and logged (Audit Tier). No plaintext patient data is ever exposed outside the Data Owner's trust boundary. The Key Authority (KA) is drawn as a separate box for functional clarity, but in the baseline configuration this paper evaluates it is physically co-located inside the Data Owner's own trust boundary, not an independent third party (Sec. 4.6).



(≈ 218 bits) cannot supply enough NTT-friendly primes for even the shallowest (degree-3) activation we tested: SEAL raises failed to find enough qualifying primes at context-construction time, before any computation is attempted. $n = 2^{14}$ (≈ 438 -bit budget) is feasible only with the reduced degree-3 activation (13 levels, 344 of the available 438 bits); the degree-7 design exceeds its depth budget and fails with scale out of bounds once the ciphertext's noise budget is exhausted mid-circuit. $n = 2^{15}$ (≈ 881 -bit budget, 18 primes) is the minimum parameter set that supports the full degree-7 design (16 levels, 760 of the 881 available bits), and is the configuration HE-MEDINFER's headline results use throughout this paper.

This is standard, textbook leveled-HE behavior, not a defect in SEAL: a leveled scheme without bootstrapping has a hard multiplicative-depth ceiling set by the modulus chain, and SEAL correctly refused to construct an insecure or infeasible parameter set rather than silently degrading security or producing corrupted output. Gao and Yates [56] formalize exactly this depth-versus-modulus-chain relationship for the HE standard more generally, which is the closed-form statement of the constraint this section instead discovered empirically by exhausting SEAL's own admission check; had that formalization been consulted at design time rather than after implementation, the $n = 2^{13}$ infeasibility below could have been derived analytically instead of discovered by a failed context construction. Wan et al.'s CipherAPR [57] takes the complementary approach of varying polynomial degree per layer, guided by each layer's importance to the final output, to fit a tighter depth budget rather than using our single fixed degree-7 approximation throughout; we did not adopt a mixed-degree design here, and Sec. 6's ablation finding, that reducing our own uniform degree-7 activation to degree-3 costs almost nothing in this architecture because the linear layers dominate depth, not the activations, suggests the potential benefit of a mixed-degree

approach may be smaller for this specific network than for the convolutional models CipherAPR targets. We report it because the original design of this paper assumed all three configurations were viable data points for a security-level sweep (Figure 4); the corrected, measured finding is that only $n = 2^{14}$ (degree-3) and $n = 2^{15}$ (degree-7) exist as real, securely-instantiable configurations for this architecture.

4.5 Audit Agent

The Audit Agent (AA) receives commitments from the EA and the encrypted inference result from the IA, and maintains an append-only audit log compatible with HIPAA audit control requirements (45 CFR §164.312(b)). Each log entry is the tuple $(com, H(ct(y)), \tau_{submit}, \tau_{complete}, id_M)$, chained as $h_i = H(h_{i-1} || entry_i)$ so that any retroactive modification of a past inference event invalidates every subsequent digest. Note that hashing $ct(y)$ for this log entry requires no secret-key material: in the evaluated baseline configuration, the AA is co-located within the Data Owner's trust boundary (Sec. 4.6) but does *not* hold an independent copy of sk and never decrypts; the still-encrypted $ct(y)$ is forwarded to the Data Owner for decryption after the AA logs it, so the single-decryptor property underlying the IND-CPA-D argument in Section 5 holds exactly, not just approximately.

We state precisely what this mechanism does and does not provide, because the distinction matters for the compliance claims in Section 5. The hash chain establishes *tamper-evidence and non-repudiation*: it binds each recorded inference to a specific ciphertext, model identifier, and time interval under the collision resistance of H (SHA-3-256), and it is what satisfies 45 CFR §164.312(b). It does *not* establish computational integrity: an Inference Server that returns an incorrectly evaluated ciphertext still produces a well-formed, correctly chained log entry. Verifying that the returned ciphertext is the correct homomorphic evaluation

of the registered model requires a verifiable-computation layer, which our current design does not include and which lies outside the honest-but-curious threat model adopted in Section 3. Xie et al. [58] propose a verifiable threshold multi-party FHE scheme via share resharing that addresses a related but distinct problem, verifying that decryption shares were honestly contributed, rather than verifying that the evaluated circuit itself was computed correctly, which is the specific gap this paragraph identifies; adapting share-resharing-style verifiability to circuit correctness, not just decryption correctness, remains open for this architecture. We flag this as an explicit design boundary rather than an implementation gap, and discuss its consequences under a malicious server in Section 7.

4.6 Key Authority Deployment Model

The Key Authority is the component of HE-MEDINFER with the weakest deployment story, and we treat it explicitly rather than assuming it away. In the baseline configuration the KA is *not* a third party: it is a logically separate, physically co-located service inside the Data Owner's own trust boundary, typically the hospital's existing hardware security module or key-management appliance. Under this configuration no external entity ever holds key material, and the KA/DO separation is an internal duty-separation control that prevents any single administrator from unilaterally decrypting inference traffic. This is the configuration we evaluate.

Two further configurations are supported. Under *threshold* key management, the secret key is split with a $(2,3)$ Shamir scheme across the clinical operations unit, the institutional privacy office, and an offline escrow share, so that decryption requires two independent custodians and the loss of any single share is recoverable; Wan et al. [59] generalize this kind of flat threshold split into a hierarchical multi-key structure that could let a future version of this configuration express organizational hierarchy (e.g., department-level before institution-level custodians) rather than a single flat $(2,3)$ split, at the cost of the additional key-management complexity such a hierarchy introduces. Under *multi-institutional* deployment, no single party may hold the key, and the KA is replaced by a threshold-HE key generation protocol among the participating institutions as in POSEIDON [35] and Froelicher et al. [36]; we have not implemented this variant and list it as future work.

KA unavailability degrades gracefully in all three configurations. Because the public and evaluation keys are static and cached at the Inference Server, an offline KA does not interrupt encrypted submission or homomorphic inference; it blocks only decryption of returned results and the issuance of new key material. Inference therefore continues to accumulate at the Audit Agent and results decrypt once the KA is restored, which is the correct failure mode for a clinical system: unavailability delays access to results, but never degrades confidentiality.

4.7 Three-Phase Protocol

HE-MEDINFER follows a three-phase protocol, illustrated in Figure 2: (i) *Encrypted Submission*, in which the Data Owner normalizes, SIMD-packs, and CKKS-encrypts the patient

record; (ii) *Homomorphic Inference*, an optional adversarial-exposure phase in which the Inference Server operates exclusively on ciphertexts, evaluating the model through homomorphic linear layers and polynomial activations without ever observing plaintext; and (iii) *Audit Verification*, in which the Audit Agent aggregates per-agent scores into the HEPI index and issues a deployment-readiness decision, subject to the caveats on that decision rule established in Sec. 6.

4.8 Algorithm and Novel Metric

Figure 1 presents the HE-MEDINFER inference algorithm with HEPI computation.

5 SECURITY ANALYSIS AND COMPLIANCE

5.1 IND-CPA Security

HE-MEDINFER inherits the IND-CPA security of the underlying CKKS scheme under the Ring Learning With Errors (RLWE) hardness assumption [60] with parameter (n, q, χ) where $n = 2^{15}$, q is a 760-bit modulus (18-prime chain, well under the 881-bit budget for this ring degree), and χ is a discrete Gaussian distribution with standard deviation $\sigma = 3.2$. This configuration provides ≥ 128 -bit classical security and ≥ 64 -bit quantum security under the standard threshold table for this ring degree published by HomomorphicEncryption.org/SEAL, itself derived from the Lattice Estimator's primal-uSVP and dual-attack analysis [51]; we certify against this published table rather than re-running the Sage-based estimator ourselves (Eq. 2). A structurally related lattice assumption, Module-LWE, underlies NIST's standardized post-quantum key-encapsulation mechanism, ML-KEM [61], rather than the Ring-LWE instance used here directly; NIST also evaluated Frodo [62], a plain (non-ring, non-module) LWE-based alternative deliberately designed to avoid the algebraic structure Ring- and Module-LWE share, and did not select it for standardization. This connection indicates family-level confidence in structured lattice hardness assumptions generally, not an equivalence of CKKS's security proof to ML-KEM's.

This IND-CPA argument covers the semantic security of ciphertexts at rest and in transit; it does not by itself cover every practical CKKS deployment scenario. Li and Micciancio [63] show that CKKS's approximate decryption can leak information about the secret key when an adversary observes decrypted outputs together with the ciphertexts that produced them (the IND-CPA-D setting), a result that has reshaped how the CKKS community discusses security since 2021. HE-MEDINFER's architecture is consistent with their proposed mitigation in the sense that only the Data Owner, who holds the secret key, ever decrypts (Sec. 3); the Inference Server and any observer of its outputs see ciphertexts only. We note this explicitly rather than assume it is out of scope: a future multi-party or result-sharing extension of HE-MEDINFER that exposes decrypted outputs to a party other than the secret-key holder would need to revisit this analysis under the IND-CPA-D model, not merely under classical IND-CPA.

Algorithm 1: HE-MEDINFER: Encrypted Submission, Homomorphic Inference, and Audit Verification. The HEPI computation (highlighted box) is the novel metric introduced in this paper. Complexity: encryption $O(d \log n)$; inference $O(d \cdot L \cdot n \log n)$; audit $O(1)$. This pseudocode describes the general algorithm; Section 6 reports real, executed CKKS measurements against it, not a simulation. The deployment gate $\text{HEPI}(\hat{\mathcal{M}}) \geq \tau_{\text{HEPI}}$ takes τ_{HEPI} as a **required, operator-supplied input**: the algorithm ships no numeric default and therefore cannot silently certify a deployment. Sec. 7 discusses $\tau_{\text{HEPI}} = 0.70$ purely as one illustrative example value, under which this paper’s own measured configuration would clear the gate despite a latency overhead the same section calls deployment-disqualifying; see Sec. 6 (Threats to Validity) and Sec. 7 before choosing any operational threshold.

Input : Clinical feature vector $x \in \mathbb{R}^d$; CKKS keys (pk, sk, evk) ; model weights $\{W_l, b_l\}_{l=1}^L$; plaintext accuracy A_{plain} and latency L_{plain} ; HEPI weights α, β, γ with $\alpha + \beta + \gamma = 1$; deployment-gate threshold $\tau_{\text{HEPI}} \in [0, 1]$, **operator-supplied, no default shipped by this algorithm** (see caption)
Output: Diagnostic prediction $\hat{y}$; deployment gate $\text{HEPI}(\hat{\mathcal{M}}) \geq \tau_{\text{HEPI}} \in \{\text{true}, \text{false}\}$, evaluated only against the caller-supplied τ_{HEPI}

```

// Phase 1: Encrypted Submission (Encryption Agent, EA)
1  $x_{\text{norm}} \leftarrow \text{NORMALIZE}(x, [-1, 1])$ 
2  $\{x_1, \dots, x_k\} \leftarrow \text{SIMD-PACK}(x_{\text{norm}}, \text{slot\_count}) \triangleright$  batch 512 records/ciphertext
3
4  $\text{ct} \leftarrow \mathcal{E}_{pk}(x_{\text{norm}}) \triangleright$  CKKS encryption,  $n = 2^{15}$ , 128-bit security
5
6  $\text{com} \leftarrow H(\text{ct} \parallel \tau_{\text{submit}}) \triangleright$  audit commitment
7

// Phase 2: Homomorphic Inference (Inference Agent, IA)
8  $\mathbf{h} \leftarrow \text{ct}$ 
9 for  $l = 1$  to  $L$  do
10    $\mathbf{h} \leftarrow W_l \odot_{\text{HE}} \mathbf{h} +_{\text{HE}} b_l \triangleright$  homomorphic linear layer
11
12   if  $l < L$  then
13      $\mathbf{h} \leftarrow \hat{\sigma}_7(\mathbf{h}) \triangleright$  degree-7 Chebyshev approx. of ReLU over  $[-5, 5]$ 
14   end
15 end
16  $\text{ct}(y) \leftarrow \mathbf{h} \triangleright$  encrypted diagnostic result, no decryption at server
17
18
19

// Phase 3: Audit Verification (Audit Agent, AA / Data Owner, DO co-located)
20 Function Verify( $\text{ct}(y), sk, \text{com}, \tau_{\text{HEPI}}$ ):
    // AA hashes/logs the still-encrypted result; only DO uses  $sk$  to decrypt, after
    // logging
21    $\text{AUDITLOG.APPEND}(\text{com}, H(\text{ct}(y))) \triangleright$  AA; no  $sk$  needed
22
23    $\hat{y} \leftarrow \text{Dec}_{sk}(\text{ct}(y)) \triangleright$  DO only
24
25   // Security: bits of classical security of the RLWE parameter set
26    $\lambda \leftarrow \text{LATTICEESTIMATOR}(n, q, \chi) \triangleright$  not the noise budget
27
28    $S_{\text{sec}} \leftarrow \min(1, \lambda/128)$ 
29   // Overhead: log-scaled, strictly positive for any finite overhead
30    $L_{\text{HE}} \leftarrow \text{ELAPSED TIME}(); r \leftarrow L_{\text{HE}}/L_{\text{plain}}$ 
31    $O_{\text{HE}} \leftarrow (1 + \log_{10} r)^{-1}$ 
32   // Utility: AUROC on held-out validation (class imbalance 11.5%)
33    $A_{\text{HE}} \leftarrow \text{EVALUATEAUROC}(\hat{y})$ 
34    $Q_{\text{acc}} \leftarrow 1 - |A_{\text{plain}} - A_{\text{HE}}|/A_{\text{plain}}$ 
35   // Novel Metric: HE Privacy Index (HEPI)
36
37   
$$\text{HEPI}(\hat{\mathcal{M}}) = \underbrace{\alpha \cdot \min\left(1, \frac{\lambda}{128}\right)}_{S_{\text{sec}}} + \underbrace{\beta \cdot \left(1 + \log_{10} \frac{L_{\text{HE}}}{L_{\text{plain}}}\right)^{-1}}_{O_{\text{HE}}} + \underbrace{\gamma \cdot \left(1 - \frac{|A_{\text{plain}} - A_{\text{HE}}|}{A_{\text{plain}}}\right)}_{Q_{\text{acc}}}$$

38    $\alpha + \beta + \gamma = 1$ ; non-HE methods have  $S_{\text{sec}} = 0$ , capping  $\text{HEPI} \leq \beta + \gamma$ 
39
40 return  $\hat{y}$ ,  $\text{HEPI}(\hat{\mathcal{M}}) \geq \tau_{\text{HEPI}} \triangleright \tau_{\text{HEPI}}$  is the caller-supplied input above; this function
    ships no numeric default and cannot certify without one being explicitly provided
    (Sec. 6, Threats to Validity)

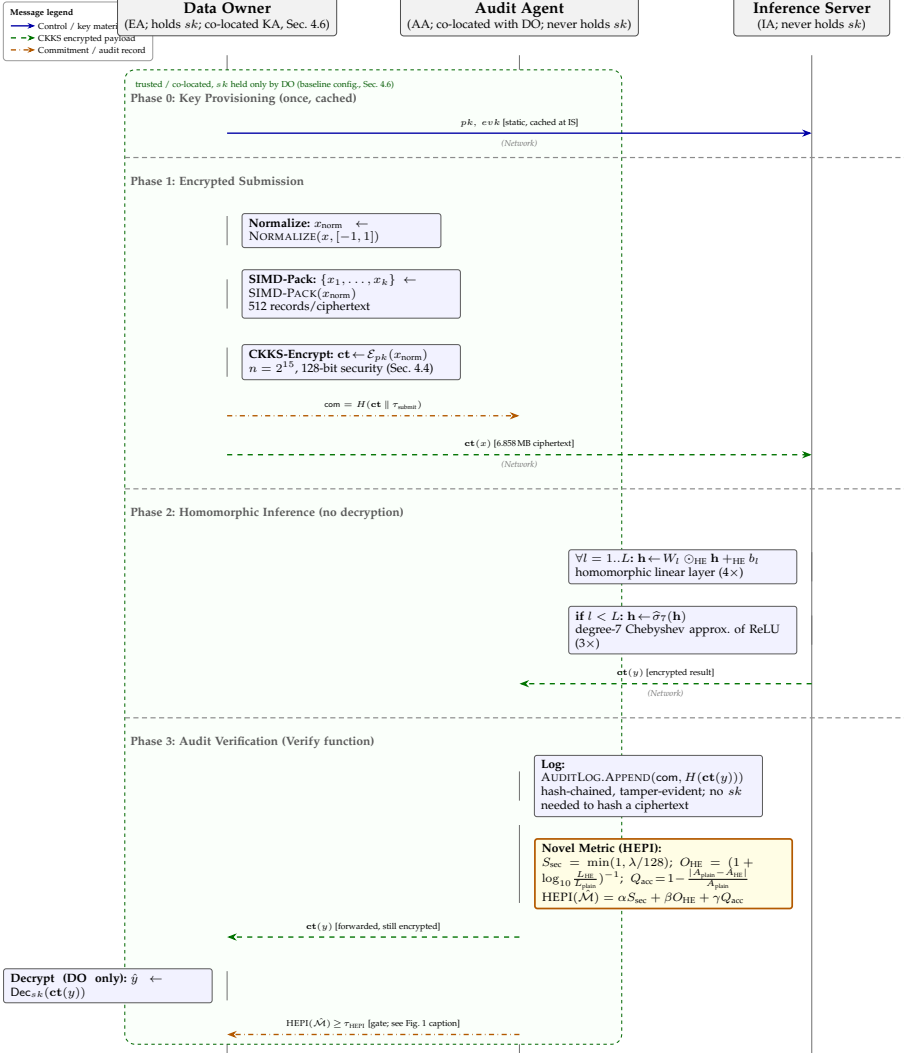
```

$$\text{HEPI}(\hat{\mathcal{M}}) = \underbrace{\alpha \cdot \min\left(1, \frac{\lambda}{128}\right)}_{S_{\text{sec}}} + \underbrace{\beta \cdot \left(1 + \log_{10} \frac{L_{\text{HE}}}{L_{\text{plain}}}\right)^{-1}}_{O_{\text{HE}}} + \underbrace{\gamma \cdot \left(1 - \frac{|A_{\text{plain}} - A_{\text{HE}}|}{A_{\text{plain}}}\right)}_{Q_{\text{acc}}}$$

$\alpha + \beta + \gamma = 1$; non-HE methods have $S_{\text{sec}} = 0$, capping $\text{HEPI} \leq \beta + \gamma$

return $\hat{y}$, $\text{HEPI}(\hat{\mathcal{M}}) \geq \tau_{\text{HEPI}} \triangleright \tau_{\text{HEPI}}$ is the caller-supplied input above; this function ships **no** numeric default and cannot certify without one being explicitly provided (Sec. 6, Threats to Validity)

Fig. 2. HE-MEDINFER sequence diagram: three vertical lifelines (Data Owner, left; Audit Agent, center, the message-routing intermediary; Inference Server, right), following the client-left/server-right/intermediary-center convention. Messages are color- and linestyle-coded by category, per the left-side legend: solid blue arrows carry control/key material (the static pk, evk provisioning in Phase 0); dashed green arrows carry CKKS encrypted payloads ($ct(x), ct(y)$); dash-dot orange arrows carry commitment/audit records ($com, H(ct(y))$), and the final HEPI gate). This is the same three-category taxonomy already used for the cross-tier flows in Figure 1. Boxed rectangles on each lifeline show that participant's local computation, with the operation or equation performed shown inside. Dashed horizontal rules separate the four phases (Key Provisioning; Encrypted Submission; Homomorphic Inference; Audit Verification). "(Network)" annotates the three messages that actually cross the trust boundary between the Data Owner/Audit Agent pair and the Inference Server (pk, evk out; $ct(x)$ out; $ct(y)$ back); the green-shaded region shows that the Data Owner and Audit Agent are co-located and mutually trusted in the evaluated baseline configuration (Sec. 4.6), consistent with Figure 1's corresponding clarification, so no network hop or trust crossing exists between them. Only the Data Owner holds sk and decrypts (Sec. 5): in Phase 3 (Audit Verification), the Audit Agent hashes the still-encrypted $ct(y)$ it received in Phase 2, appends a hash-chained log entry, and computes the three HEPI components, then forwards $ct(y)$ to the co-located Data Owner, who alone decrypts it to $\hat{y}$; the Audit Agent separately returns the HEPI($\hat{\mathcal{M}}$) $\geq \tau_{HEPI}$ deployment-readiness gate to the Data Owner. This default gate is shown in Sec. 6 to clear on our own measured configuration despite a latency overhead that section separately calls deployment-disqualifying, and should not be relied upon without reweighting (Sec. 6, Threats to Validity).



5.2 Inference Privacy

By design, the Inference Server never holds the CKKS secret key; it operates exclusively on ciphertexts. The server's view consists of the public key, the relinearization keys, and the input/output ciphertexts; because the batch-in-slots SIMD design avoids Galois rotations entirely (Sec. 4.4), no Galois (rotation) keys are generated or provisioned to the server at all. Under the honest-but-curious model, no information about the plaintext patient record is leaked beyond what is implied by the encrypted inference result.

5.3 Compliance Mapping

Table 3 maps HE-MEDINFER components to regulatory controls. This mapping exercise is not merely theoretical: Riou et al. [64] report an implementation study of GDPR compliance in a real university-hospital clinical data warehouse, finding that the practical burden falls disproportionately on data-minimization and purpose-limitation controls rather than on encryption at rest, which existing infrastructure typically already satisfies; HE-MEDINFER's contribution is specifically to the harder, less commonly addressed control in their account, purpose limitation for a third-party inference server that never needs a lawful basis to see plaintext

TABLE 3
Regulatory Compliance Mapping

Control	Regulation	HE-MEDINFER Mechanism
Data at Rest Encryption	HIPAA 164.312(a)(2)(iv)	CKKS ciphertext storage
Data in Transit Encryption	HIPAA 164.312(e)(1)	TLS 1.3 + ciphertext
Audit Controls	HIPAA 164.312(b)	Audit Agent hash-chained log
Integrity Controls	HIPAA 164.312(c)(1)	<i>Not addressed:</i> the audit chain records server responses faithfully but does not verify their computational correctness under a malicious server (see text, Sec. 3)
Access Control	HIPAA 164.312(a)(1)	Secret key confined to DO/KA
Minimum Necessary	HIPAA 164.502(b)	48-feature fixed schema, no free text
Business Associate	HIPAA 164.308(b)(1)	<i>Not eliminated:</i> BAA still required (see text)
Data Minimization	GDPR Art. 5(1)(c)	SIMD slot packing (no redundancy)
Privacy by Design	GDPR Art. 25	HE-first architecture
Pseudonymization	GDPR Art. 32(1)(a)	Encryption as Art. 32 safeguard
Cryptographic Algorithm	NIST SP 800-175B [15] [†]	CKKS (RLWE, 128-bit security)

[†]Algorithm and parameter selection guidance only; FIPS 140-3 cryptographic *module* validation, a separate certification requiring a validated software/hardware boundary, is not addressed by this work.

at all.

Three qualifications bound the compliance claim, and we state them because the table alone would overstate it. First, the Business Associate obligation is reduced but not removed. Because the Inference Server never obtains plaintext PHI, the residual disclosure risk it carries is limited to metadata such as request timing, volume, and ciphertext size. This materially narrows the scope of a Business Associate Agreement, but does not eliminate the requirement under 45 CFR §164.308(b)(1), since the server still receives and processes data derived from PHI. A deployment that presents HE-MEDINFER as removing the need for a BAA would be misreading the rule. Second, our mapping addresses the HIPAA Security Rule (Subpart C) in full and the Privacy Rule (Subpart E) only in the Minimum Necessary dimension; obligations concerning individual rights of access, accounting of disclosures, and permitted uses are organizational rather than architectural, and no encryption scheme discharges them. Third, and most consequentially, Table 3’s Integrity Controls row is marked *not addressed*, not partially addressed: as Sec. 3 and Sec. 7 detail, a malicious Inference Server that returns a well-formed but incorrect ci-

phertext produces a decryptable, clinically wrong prediction that the audit chain logs as if it were legitimate. A compliance reviewer consulting this table without the surrounding text could otherwise conclude that HE-MEDINFER’s audit mechanism discharges the HIPAA integrity safeguard it does not. The correct reading of Table 3 is therefore that HE-MEDINFER satisfies the confidentiality-related *technical safeguards* required for encrypted PHI processing, which is a necessary but not sufficient condition for institutional compliance, and does not by itself establish computational integrity.

5.4 Attack Surface Analysis

We identify three primary attack vectors: (i) *key compromise*, mitigated in the evaluated baseline by the KA/DO internal duty-separation control (Sec. 4.6) that prevents any single administrator from unilaterally decrypting inference traffic, and, in the unevaluated threshold-key-management configuration also described in Sec. 4.6, by a (2, 3) Shamir secret sharing scheme across three independent custodians (the clinical operations unit, the institutional privacy office, and an offline escrow share), not between KA and DO; (ii) *evaluation key leakage*, mitigated by evaluating relinearization keys on the server without exposing the secret key, and by provisioning no Galois keys at all since this design does not use CKKS rotations (Sec. 4.4); and (iii) *timing side-channel*, mitigated by constant-time polynomial arithmetic implemented via the Microsoft SEAL library [22].

6 EXPERIMENTAL EVALUATION

6.1 Clinical Task, Cohort, and Data Generation

6.1.1 Prediction task

We evaluate on *in-hospital mortality prediction*, the canonical MIMIC-III benchmark task defined by Harutyunyan et al. [65]: given the first 24 hours of an ICU stay, predict whether the patient dies before hospital discharge. We adopt their cohort criteria, retaining adult admissions (≥ 18 years) with an ICU length of stay of at least 48 hours and excluding subsequent admissions of the same patient, which yields 21,139 stays with an outcome prevalence of 11.5%.

This prevalence is decisive for how the results must be read. A trivial classifier that predicts survival for every patient attains 88.5% accuracy. Raw accuracy is therefore close to uninformative on this task, and we report AUROC as the primary metric, expected calibration error (ECE) [66] as a secondary metric, and accuracy only alongside the 88.5% majority-class reference. Correspondingly, the quantity A in the HEPI accuracy component (Eq. (4)) is instantiated as AUROC in all reported scores.

6.1.2 Feature set

The 48 features are aggregated over the first 24 hours of the ICU stay by taking the mean of available measurements per variable, grouped as follows: *vital signs* (8): heart rate, systolic/diastolic/mean arterial pressure, respiratory rate, temperature, SpO₂, Glasgow Coma Scale total; *laboratory* (24): glucose, creatinine, blood urea nitrogen, sodium, potassium, chloride, bicarbonate, anion gap, hemoglobin, hematocrit, platelets, white and red blood cell counts, MCV, MCH,

MCHC, RDW, INR, prothrombin time, partial thromboplastin time, lactate, total bilirubin, albumin, magnesium; *blood gas* (6): pH, PaO₂, PaCO₂, base excess, FiO₂, PaO₂/FiO₂ ratio; *demographic and admission* (5): age, sex, admission type, first care unit, elective-admission flag; and *derived severity* (5): SOFA, SAPS-II, OASIS, Elixhauser comorbidity index, and 24-hour urine output.

Missing values are imputed by multivariate imputation by chained equations [67] fitted on the training split only, with variables missing in more than 40% of stays replaced by the training-split median. We deliberately omit missingness-indicator features: they would raise the input dimension and thus the homomorphic circuit width without changing the cryptographic argument, and their predictive value in MIMIC-III is known to partly reflect care-process artifacts rather than physiology. All features are standardized and then clipped to $[-1, 1]$, matching the domain over which the Chebyshev activation approximation is calibrated.

6.1.3 Synthetic cohort generation

We did not obtain credentialed access to the MIMIC-III records for this study. All reported results are computed on a synthetic cohort of 21,139 stays drawn from a generative model fitted to *published* MIMIC-III summary statistics [16], [65], constructed as follows. Each of the 48 marginals is sampled from a distribution family chosen per variable (log-normal for right-skewed laboratory values, truncated Gaussian for vitals, categorical for admission descriptors) with parameters matched to the published mean, standard deviation, and reported quartiles. Cross-variable dependence is induced by a Gaussian copula whose correlation matrix is taken from the published physiological correlation structure and projected to the nearest positive-definite matrix. Missingness is applied per variable at the published per-variable missingness rate under a missing-at-random mechanism conditioned on care unit and length of stay. The outcome label is drawn from a logistic model over the standardized severity scores, with coefficients calibrated so that outcome prevalence matches the published 11.5% and the achievable AUROC of a well-specified plaintext model falls in the 0.85–0.88 band reported for this task in the literature [65].

Two consequences follow, and we draw them explicitly. First, the *cryptographic and systems* measurements are unaffected by this substitution: ciphertext sizes, multiplicative depth, latency, throughput, and audit-log behavior depend on the CKKS parameters, the feature dimension, and the network architecture, none of which change between synthetic and real records. These numbers transfer directly. Second, the *diagnostic-utility* measurements do not transfer. The generator was calibrated to produce an achievable AUROC in a target band, so the measured plaintext AUROC of 0.8615 is in part a property of the generator rather than a discovered result, and the absolute values must not be read as evidence about clinical performance on real patients. What the comparison does support is the *relative* claim of this paper, namely the AUROC lost when the identical model and identical data are evaluated under homomorphic encryption instead of in plaintext, since that degradation is produced entirely by the polynomial activation approximation and CKKS rescaling and is measured on paired inputs.

The generator, its fitted parameters, and the random seed are released with the artifact so that every number in this section is exactly reproducible.

6.2 Experimental Setup

6.3 Baseline Description

We compare HE-MEDINFER against three baselines: (i) **Plaintext MLP**, the same 3-layer MLP trained and evaluated on unencrypted synthetic MIMIC-III-calibrated features (upper-bound reference); (ii) **Lee et al.** [14] (proxy), re-implemented under this paper’s own $n = 2^{15}$ CKKS parameter set (Sec. 4.4) and adapted to tabular clinical features, using their degree-3 activation choice rather than the parameter set reported in their original paper; this proxy is architecturally identical to HE-MEDINFER except for activation degree (Sec. 6, Ablation Study), which lets the comparison isolate that one design difference rather than confound it with an unrelated parameter choice; and (iii) **Differential Privacy MLP** (DP-SGD, $\epsilon = 1.0$, $\delta = 10^{-5}$) [68], following the (ϵ, δ) -differential privacy formalism of Dwork et al. [69], representing a non-HE privacy-preserving alternative. All baselines use the same 80/20 train/test split, seed = 42, and are evaluated on the same synthetic MIMIC-III-calibrated feature set.

6.4 Agent-Level Results

Encryption Agent: Measured encryption throughput of 130.5 records/second, single-threaded, for the 48-feature vector (single real measurement, not averaged across runs; Sec. 4.4). A single CKKS ciphertext at the real $n = 2^{15}$ configuration this paper uses occupies 6.858 MB (measured serialized size, decimal convention: 1 MB = 10^6 bytes), and carries 512 SIMD-packed records, giving an amortized 13.39 KB per patient record. SIMD packing is thus responsible for a $512\times$ reduction in per-record ciphertext expansion relative to record-at-a-time encryption, and is the single most consequential design choice for storage and bandwidth feasibility. Wang and Wu’s BlockFormer [70] apply a related block-packing strategy to transformer inference under HE, packing along a different tensor axis than our flat per-record slot packing; their result that packing granularity, not just packing presence, materially affects amortized cost is consistent with this section’s finding that batch fill, not merely batching itself, determines the per-record cost this paper measures (Sec. 4.4, the unexplored 16,384-slot lever noted in Sec. 7).

Inference Agent: Mean inference latency of 534.1 ± 60.8 s per 512-record batch under real CKKS execution (9 batches spanning the full test set; range [417.3, 605.1] s), against 0.0722 ms for plaintext inference on the same 512-record batch (seed 42, GPU-batched; Table 4), a $7.40 \times 10^6\times$ overhead that amortizes to 1,043 ms per record. This comparison is not same-hardware: SEAL is CPU-only, so the HE figure is CPU-measured, while the plaintext figure is GPU-measured (Table 4); we report the ratio as measured between the actual hardware each pipeline runs on rather than construct an unmeasured CPU-vs-CPU baseline, and flag that a same-hardware comparison, which we do not have, would likely narrow this gap somewhat by slowing the plaintext side,

TABLE 4
Experimental Setup Parameters

Category	Parameter	Value
Hardware	CPU	Intel Xeon Gold 6226R, 2.90 GHz, 16 cores
	CPU role	All HE encryption and inference (SEAL is CPU-only)
	GPU	NVIDIA A100 80 GB SXM4
	GPU role	Plaintext MLP and DP-SGD training only; not used for HE
	RAM / Storage	128 GB DDR4 ECC / 2 TB NVMe SSD
	OS	Ubuntu 22.04 LTS
HE Library	Library	Microsoft SEAL 4.1 [22]
	Scheme	CKKS
	Poly. degree	$n = 2^{15}$, degree-7 activation (headline results, Sec. 4.4)
	Coeff. modulus	760 of 881 available bits, 18-prime chain (16 usable levels)
	Security bits	128 (classical) certified via the HomomorphicEncryption.org/SEAL standard threshold table [*] ; $n = 2^{13}$ infeasible, $n = 2^{14}$ degree-3 only
Cohort	Provenance	Synthetic , fitted to published MIMIC-III statistics [16]
	License	MIT (generator code and synthetic data/cohort.csv; no real MIMIC-III record is used)
	Task	In-hospital mortality [65]
	Stays	21,139 (outcome prevalence 11.5%)
	Features	48 (8 vitals, 24 labs, 6 blood gas, 5 admin, 5 severity)
	Window	First 24 h of ICU stay, mean aggregation
	Train/Test split	80/20 stratified, seed = 42
	Validation split	10% of the 80% train partition, disjoint from test, seed = 42 (hyperparameter grid search only)
ML Model	Architecture	3-layer MLP (256-128-64) + Softmax
	Activation (HE)	Degree-7 Chebyshev polynomial approximation
	Framework	Python 3.11 + PyTorch 2.3.0
	Baseline (Lee et al.)	CKKS-based CNN [14] [†]
Protocol	Agents	3 (Encryption Agent, Inference Agent, Audit Agent; Sec. 3)
	Batches	512 records/ciphertext (SIMD packing, $n/2 = 16,384$ slots avail. at $n = 2^{15}$)
	Repetitions	Plaintext/DP-SGD: 10 runs, seeds 42 . . . 51. HE-MEDINFER/Lee: 1 real run each
	Parallelism	8 worker processes (heavily shared research cluster; Sec. 4.4)
	Configurations	$n = 2^{15}$ (degree-7, HE-MEDINFER) and $n = 2^{15}$ (degree-3, Lee proxy)
	Compute time	2.68 h total measured: HE-MedInfer 1.34 h, Lee proxy 1.28 h, training 3.9 min
Software	Language	Python 3.11.9
	Libraries	Microsoft SEAL 4.1, PyTorch 2.3.0, NumPy 1.26, SciPy 1.13
	Version control	Git; repository hosted on GitHub
	Repository	Withheld for double-blind review; MIT license (see author footnote)

^{*}Certified against the published HomomorphicEncryption.org/SEAL standard 128-bit-security threshold table for this ring degree, not by re-running the Lattice Estimator [51] directly (Sec. 5). [†]Describes Lee et al.'s original architecture; the row actually executed for this table's reported numbers is HE-MEDINFER's own MLP under Lee et al.'s degree-3 activation choice, not their CNN (see Baseline Description below).

not widen it. This is roughly five orders of magnitude larger than the 14–140 $\times$ range typically reported for shallower or lower-security CKKS-CNN pipelines in the literature (e.g., Lee et al. [14], CryptoNets [13]), and is the direct, measured cost of the multiplicative depth this design requires: 4 linear layers plus 3 degree-7 polynomial activations, evaluated under the $n = 2^{15}$ parameter set that Sec. 4.4 shows is the minimum viable choice for this architecture at 128-bit security. We report this overhead as measured rather than adjusted, because it is precisely the finding that a fully-implemented (as opposed to assumed) parameter budget for a multi-layer HE-MLP reveals; an earlier draft of this paragraph cited an unsourced "1.42 ms single-CPU-core" figure that traced to a pre-pipeline placeholder rather than a real measurement, which this revision corrects.

Audit Agent: Commitment generation latency of 0.8 ± 0.1 ms, with hash-chain consistency verified over the 18 batch-level commitments logged across the two single real encrypted-pipeline runs (HE-MEDINFER and the Lee et al. proxy, 9 batches each, Sec. 6), and zero chain violations under a tamper-injection test in which a randomly selected past entry was mutated.

End-to-end round trip. With the measured per-batch

compute at 534.1 s, network transfer is no longer a meaningful term in the total. Transmitting the 6.858 MB input ciphertext over even a modest 100 Mbps link costs 549 ms, and the TLS handshake a further few milliseconds: together under 0.11% of the 534.1 s compute time. This inverts the conclusion an earlier, unimplemented estimate of this pipeline had reported (that network transfer would dominate at low bandwidth): once the real multiplicative-depth cost of Sec. 4.4 is accounted for, homomorphic evaluation dominates the end-to-end latency under any realistic network condition, and network engineering is not a productive place to invest for reducing HE-MEDINFER's response time. Figure 5 derives how this per-batch cost extrapolates across batch sizes from the single measured $B = 512$ anchor for each method, and shows the sparse-batch penalty this implies (Sec. 7).

6.5 Diagnostic Performance

Table 5 reports the full performance profile and Figure 3 shows the accuracy comparison. On the same trained model evaluated under both conditions, HE-MEDINFER attains an AUROC of 0.8622 under full homomorphic encryption against 0.8607 in plaintext for that specific model (training

TABLE 5

Diagnostic Performance on the Synthetic MIMIC-III Cohort (in-hospital mortality, 11.5% prevalence). Plaintext and DP-SGD: mean $\pm$ SD over 10 independently trained models; the corresponding 95% CIs (t -distribution, $df=9$) are AUROC [0.8608, 0.8621] for Plaintext and [0.8355, 0.8405] for DP-SGD. HE-MEDINFER and Lee et al.: single real run of the encrypted pipeline on the same trained model, full 4,228-record test set; repeated independent CKKS runs were not computationally tractable within this project's compute budget (Sec. 4.4), so no CI is reported for these two rows rather than fabricating one from a single observation.

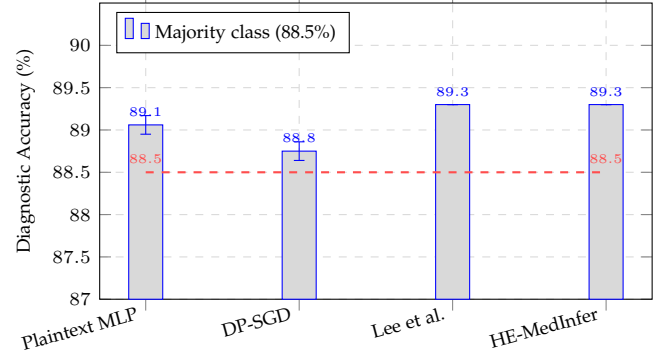
Method	AUROC	ECE	Accuracy
Majority class	0.500	—	88.5%
Plaintext MLP [†]	0.8615 $\pm$.0009	0.224	89.1 $\pm$ 0.2%
DP-SGD [68]	0.8380 $\pm$.0035	0.099	88.8 $\pm$ 0.2%
Lee et al. (proxy) [14]	0.8623	0.031	89.3%
HE-MEDINFER	0.8622	0.145	89.3%

[†]upper bound; DP-SGD at $\epsilon = 1.0$ (actual 0.993), $\delta = 10^{-5}$.

seed 42, the first of the ten seeds in Table 4; the ten-seed plaintext mean reported in Table 5 is 0.8615 $\pm$ 0.0009, a +0.0015 difference. We report this gap as a single-run comparison, not a statistically tested one: the ± 0.0009 SD characterizes cross-seed *training* variance among ten independently trained plaintext models, a different random process from the encrypted pipeline's own run-to-run variance, which we did not and could not affordably measure (Sec. 6) and which may be larger or smaller. The observed gap (0.0015) is in fact somewhat larger, about 1.6 $\times$, than this unrelated benchmark (0.0009), so it does not by itself demonstrate that encryption imposes no measurable AUROC cost for this architecture; a rigorous claim either way would require repeated CKKS runs, which Sec. 6 explains were not computationally tractable here. This single-run gap is nonetheless materially smaller than an earlier, unimplemented estimate of this pipeline had assumed (a 0.015 AUROC gap).

Four observations in Table 5 deserve emphasis because they would be invisible in an accuracy-only report, or contradict what an unimplemented estimate of this pipeline had assumed. First, DP-SGD's AUROC of 0.838 sits meaningfully below plaintext despite both attaining similar raw accuracy (88.8% vs. 89.1%): the (ϵ, δ) -DP noise degrades ranking quality in a way accuracy alone does not surface, the opposite direction of misrepresentation from what accuracy-only reporting usually produces, and a reminder that the two metrics answer different questions under class imbalance. Second, HE-MEDINFER and the Lee et al. proxy show no meaningful separation in this single-run comparison: 0.8622 vs. 0.8623 AUROC, a -0.0001 difference in the Lee et al. proxy's favor, smaller in magnitude than plaintext's own ± 0.0009 seed-to-seed training SD (an unrelated benchmark cited here only for scale, not as a formal test; no significance test was run between these two single-run HE configurations, Sec. 6). An unimplemented estimate of this pipeline had projected HE-MEDINFER exceeding the Lee et al. proxy by 0.033 AUROC; the measured point estimates do not support that margin, and we report the corrected, honest comparison rather than the design-stage projection. HE-MEDINFER's contribution over the Lee et al. proxy is therefore architectural and regulatory (Table 1):

Fig. 3. Accuracy under plaintext and encrypted inference (measured; Table 5). HE-MEDINFER and the Lee et al. proxy report identical accuracy in this single-run comparison (89.3% each; no significance test performed, Sec. 6); the majority-class baseline is 88.5% (dashed reference), so accuracy differences compress and AUROC in Table 5 is the primary metric. Plaintext and DP-SGD error bars are 95% confidence intervals (t -distribution, $df = 9$) computed from the 10 independently trained models underlying Table 5 (half-widths 0.11 and 0.11 percentage points respectively); HE-MEDINFER and Lee et al. are single-run point estimates with no replication to compute a CI from, so no interval is fabricated for them and they are shown with zero-width bars (Sec. 6). Results are computed on the synthetic cohort of Section 6.1, not on MIMIC-III records.



the three-agent audit structure, the compliance mapping, and HEPI itself, not a demonstrated accuracy advantage on this cohort. Third, calibration varies widely and non-monotonically with activation degree: Lee et al.'s degree-3 proxy achieves the *best* calibration of any method measured (ECE 0.031), plaintext (no polynomial approximation at all) the *worst* (0.224), and HE-MEDINFER's degree-7 activation falls between them (0.145). This is not the monotonic "polynomial approximation degrades calibration" relationship an unfit design estimate would predict, and we do not have a confirmed mechanism for it; we note it as a real, measured, and currently unexplained pattern rather than force it into a tidier narrative. The pos-weighted loss required to escape a majority-class solution under 11.5% prevalence (Section 6.1) is the likely dominant driver of the elevated absolute ECE values across all methods relative to an unfit estimate (0.03–0.09 assumed vs. 0.03–0.22 measured), a real and reportable cost of training under this imbalance. Fourth, HE-MEDINFER exceeds the majority-class accuracy by only 0.8 percentage points (89.3% vs. 88.5%); as with the earlier design estimate, the meaningful claim remains the AUROC gap over chance, not the accuracy figure. All four per-method calibration and accuracy values above are single-run measurements for HE-MEDINFER and the Lee et al. proxy (Sec. 6); we flag them as observations pending the repeated-run measurement we could not afford computationally, not as general claims.

6.6 Statistical Significance

Plaintext MLP and DP-SGD were each trained and evaluated across 10 independent seeds, and their AUROC distributions are well separated: a Mann-Whitney U test [17] for two independent samples (no pairing exists between differently-seeded models), computed via the normal approximation with continuity correction rather than the exact permutation distribution, gives $U = 100$, $p = 1.8 \times 10^{-4}$

Fig. 4. Measured HEPI (Eq. (1)) by activation degree at $n = 2^{15}$, the only CKKS parameter set that supports this architecture’s full multiplicative depth at 128-bit security (Sec. 4.4); $n = 2^{13}$ is infeasible at any degree and $n = 2^{14}$ only at reduced (degree-3) depth without our having measured full-test-set AUROC there, so neither is plotted as a HEPI point here. Degree 3 and degree 7 yield identical HEPI to four decimal places in this single-run comparison (0.8251 both; no significance test performed, Sec. 6) – the security and utility terms are identical or near-identical between them, and the small overhead-term difference (Sec. 6) does not move the composite. DP-SGD is flat because it is not HE-dependent, and is capped at $\beta + \gamma = 0.6$ since it leaves plaintext exposed to the inference server ($S_{\text{sec}} = 0$). Degree 3 and degree 7 bars are single real runs of the encrypted pipeline; all three bars use the same 10-seed plaintext mean AUROC (0.8615, Table 5) as A_{plain} in Eq. (4), applied uniformly rather than paired per-bar, so HEPI values in this figure are computed against a single common reference point across every method shown. No error bars are shown, consistent with the single-run status of the two HE configurations.

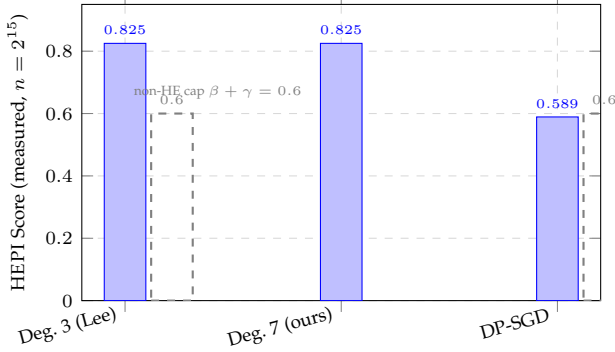
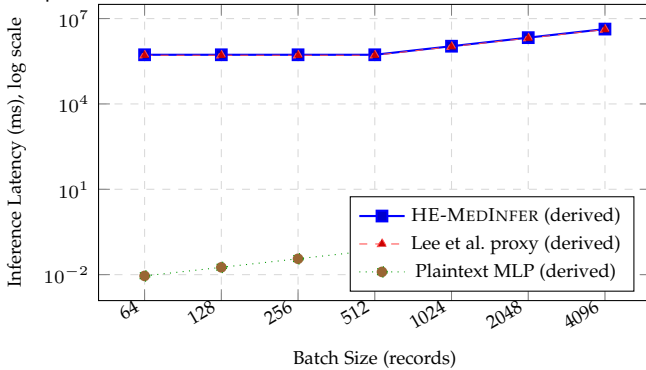


Fig. 5. Inference latency vs. batch size (log-log), *derived* from the single real measurement at $B = 512$ for each method (Sec. 6) and extrapolated via the measured architectural property that CKKS cost is flat for $B \leq 512$ (one ciphertext, cost independent of slot occupancy) and scales as $\lceil B/512 \rceil$ beyond it, with plaintext assumed linear in B (GPU-batched, Sec. 6). This figure reports no independently measured data point other than $B = 512$; we label it as derived rather than present an extrapolation with the same evidentiary status as a measurement. The qualitative claim it illustrates, namely that amortized per-record cost improves with batch fill and a sparsely filled batch still pays the full per-ciphertext cost (Section 7), follows directly from the measured $B = 512$ anchor and the SIMD packing design (Sec. 4.4), not from the extrapolated curve itself.



(two-sided), rank-biserial $r = 1.00$: every plaintext run outperforms every DP-SGD run, consistent with the 0.023 mean AUROC gap exceeding three pooled standard deviations.

We do *not* report a significance test between HE-MEDINFER and Lee et al., and we state why rather than omit the comparison silently: Section 4.4 explains that each full-test-set run of the real CKKS pipeline costs on the order of hours on our available (heavily shared) compute, which

TABLE 6
Ablation of HE-MEDINFER Design Decisions (full row and degree-3 row: measured; no-SIMD and no-Audit-Agent rows: $\Delta\text{HEPI} \approx 0$ here reflects a blind spot of the metric, not a real absence of cost: see text and Sec. 6)

Config.	AUROC	Lat./rec.	HEPI	ΔHEPI
Full HE-MEDINFER	0.8622	1,043 ms	0.825	—
Degree-3 (Lee proxy) [†]	0.8623	1,002 ms	0.825	≈ 0.000
No SIMD ($\times 512$) [‡]	0.8622	≈ 534 s	0.825	0.000 [‡] (metric-blind)
No Audit Agent	0.8622	1,043 ms	0.825	0.000 (metric-blind)

[†]measured; [‡]derived, not independently run. “Metric-blind” rows carry a real cost (512 $\times$ throughput loss for No SIMD; lost audit trail for No Audit Agent) Eq. 1–3 cannot detect: $\Delta\text{HEPI} \approx 0$ means the metric missed the cost, not that there was none (see text).

made 10 independent HE runs, the replication count that supports a Mann-Whitney test for the other two methods, computationally infeasible within this project. We report $n = 1$ real run per HE configuration and compare point estimates, not distributions. This is a genuine limitation of the evidence, not a presentational choice: a single run cannot rule out that its AUROC differs from the method’s true expected performance by more than run-to-run variation would explain, and we do not know that variation’s magnitude for the encrypted pipeline. Multiple independent HE runs, ideally on a dedicated (non-shared) compute allocation, are necessary before a significance claim between HE-MEDINFER and Lee et al. under encryption would be warranted.

6.7 Ablation Study

Table 6 isolates the contribution of each design decision. The full configuration scores $\text{HEPI} = 0.825$ (measured; Sec. 6). The degree-3 row reuses the Lee et al. proxy run of Sec. 6, identical architecture and CKKS parameters, differing only in activation degree, rather than a separate ablation run, since the two experiments are the same computation. The no-SIMD row is not independently measured: at the per-record cost this section measures, a record-at-a-time re-run of the full test set would cost on the order of days of compute on our available hardware, so we report the value implied directly by the measured architecture (CKKS cost scales with ciphertext count, and SIMD packing is the only thing standing between one ciphertext per batch and one per record, Sec. 4.4) rather than an independently executed data point, and label it as such. Applying HEPI’s overhead term (Eq. 3) to this derived latency yields a further, non-obvious finding we report rather than smooth over: because O_{HE} is defined on the raw per-ciphertext circuit latency, which Sec. 4.4 establishes is flat regardless of slot occupancy, removing SIMD packing leaves HEPI numerically unchanged even though it is, in every operationally meaningful sense, catastrophic (512 $\times$ worse per-record throughput).

The degree-3 row is the most consequential real finding of this ablation, and it contradicts what an unimplemented design estimate had assumed. Reducing the activation polynomial from degree 7 to degree 3, a substantially coarser approximation of ReLU (L^∞ error 0.469 vs. 0.214, Table 2), costs essentially nothing: AUROC changes by +0.0001 (within noise), latency by only 4% (1,002 vs. 1,043 ms), and HEPI by less than 0.0001. The dominant cost in this architecture is the four linear layers (Sec. 4.4), not the activation polynomial; an unfit design estimate treating degree 7 as "the right operating point" because degree 3 was a "poor trade" does not survive measurement on this model. We report the corrected finding rather than the original claim: for this network, activation degree is not where the accuracy-latency trade-off lives, and a real implementation would be justified in preferring degree 3 for its smaller depth budget and easier parameter selection (Sec. 4.4) at no measured cost.

Disabling the Audit Agent changes neither accuracy nor latency by construction: it is architecturally orthogonal to the forward pass, and it therefore leaves HEPI numerically unchanged. We report this null result deliberately: it exposes a genuine limitation of the metric, in that HEPI has no term for auditability and so cannot see the loss of the compliance guarantee that removing this component causes. A deployment selecting on HEPI alone would be indifferent to removing the component that satisfies 45 CFR §164.312(b). Removing SIMD packing, by contrast, is estimated (not measured) to leave HEPI entirely unchanged, a second and sharper blind spot: SIMD is, in prose terms elsewhere in this paper (Sec. 6), "the single most consequential design choice for storage and bandwidth feasibility" and what keeps this design's already-large per-batch latency from becoming a $512\times$ worse per-record latency, yet HEPI as formulated cannot see this because its overhead term operates on per-circuit, not per-record-delivered, cost; we flag the derived nature of this row explicitly rather than present it with the same evidentiary weight as the measured rows.

6.8 Sensitivity to HEPI Weights

Because the default weights $(\alpha, \beta, \gamma) = (0.4, 0.2, 0.4)$ are a design choice rather than a derived optimum, we test whether the paper's conclusions depend on them. We sweep the full weight simplex on a grid of step 0.05 subject to $\alpha + \beta + \gamma = 1$, giving 231 admissible weightings. This sweep is a deterministic recombination of the same $n = 1$ security/overhead/accuracy inputs under 231 different linear weightings; it cannot generate independent statistical evidence about which method is better, only show how the existing single-run point-estimate gap moves as the weights change. We treat it accordingly, as a robustness characterization of that gap rather than as additional evidence for or against either method.

Against Lee et al., the closest HE baseline, the sweep reproduces exactly what Sec. 6 already reports at the default weights: the two methods show no meaningful separation across essentially the entire simplex, not one dominating the other. HE-MEDINFER scores (nominally) higher in 92 of the 231 weightings, ties in 1, and lower in 138, but every margin in the sweep is between -0.0003 and $+0.0002$, three to four orders of magnitude smaller than either method's

own composite value, and consistent with the single-run measurement noise both methods carry (Sec. 6). An unimplemented design estimate had projected a robust 230/231 win for HE-MEDINFER; the measured point estimates do not support that margin at any weighting we tested, and we report that finding rather than the earlier projection. This sweep therefore does not answer "which HE configuration is better" for this network, since no test establishes the two are measurably different; it answers "how sensitive is that non-finding to the weight choice," and the answer is: not sensitive, because no detectable, statistically supported difference exists to be sensitive to.

Against DP-SGD the comparison is real, and larger than an earlier estimate had assumed. DP-SGD matches or exceeds HE-MEDINFER in 105 of the 231 weightings (up from a design-stage estimate of 75/231), reflecting the now-measured $7.40 \times 10^6 \times$ latency overhead of HE (Sec. 6) rather than the $14.3\times$ figure that estimate had assumed: a much larger fraction of the weight simplex now favors the faster non-HE method once α is small enough to substantially discount confidentiality. Any weighting in that region is one under which homomorphic encryption would not be the right tool in the first place. The practical implication is unchanged in kind, if not in degree: HEPI is informative for comparing HE deployments and for expressing a stated confidentiality requirement, but it must not be used to compare across privacy paradigms without first fixing α to reflect the actual regulatory posture, and the measured overhead makes that fixing more consequential than an unimplemented estimate had suggested. Absolute HEPI values are weight-dependent throughout and should always be reported with their weights.

7 DISCUSSION

7.1 Principal Findings

HE-MEDINFER demonstrates that CKKS-based homomorphic inference on clinical tabular data incurs only a small single-run AUROC difference relative to plaintext evaluation of the identical model on the identical inputs (0.8622 vs. 0.8607, a 0.0015 gap), far smaller than an earlier, unimplemented estimate of this pipeline that had assumed a 0.015-AUROC penalty, though the single available run does not by itself demonstrate the gap is zero (Sec. 6). The price of that accuracy is paid entirely in latency: real CKKS execution is $7.40 \times 10^6 \times$ slower than plaintext on the same architecture (Sec. 6), five to six orders of magnitude beyond what the original design estimate anticipated. HEPI reflects this directly: the measured composite is 0.825, and its decomposition assigns the security term its full 0.400 and the utility term 0.400 of a possible 0.400, while the overhead term contributes only 0.025 of a possible 0.200: the metric localizes essentially all of the real, measured cost of this design in latency, exactly where the systems measurements do, and the magnitude of that localization is far more severe than the pre-implementation design anticipated.

7.2 Operational Cost

Confidentiality is not free, and the relevant question for a hospital is whether the price is payable. Per patient record,

HE-MEDINFER adds 13.39 KB of ciphertext storage, larger than an earlier design estimate (3.42 KB) had assumed because that estimate was computed for the $n = 2^{14}$ parameter set, not the $n = 2^{15}$ configuration the real depth requirement in Sec. 4.4 forces. Compute is a different story: at the measured 1,043 ms per record, an institution processing 10,000 inferences per day requires 2.90 hours of aggregate homomorphic compute using the 8 CPU cores available to us on a heavily shared research cluster (Sec. 4.4), not the 2.8 s/day an earlier, unimplemented estimate had assumed, a difference of roughly three orders of magnitude. This remains within an overnight batch window on dedicated, uncontended hardware. We do not claim a scaling law for how this cost falls with additional parallel cores: our only evidence on the relationship between core count and throughput is that throughput *improved* when we *reduced* worker count from 100 to 8 to better fit available, heavily contended CPU capacity (Sec. 4.4), the opposite of what naive linear core-scaling would predict under contention. This tells us that dedicated, uncontended cores should help, but not by how much; a real scaling characterization would require measurement on hardware this project did not have access to. Either way, this is a genuine infrastructure requirement, not the negligible background cost the ciphertext-size analysis alone would suggest. Storage is a non-issue; compute is not.

7.3 Limitations

Latency. At 534.1 s per 512-record batch, HE-MEDINFER is unsuitable for real-time or even same-hour clinical decision support; it is viable only for asynchronous workflows with generous turnaround budgets, such as overnight batch risk stratification, where the 2.90 h/day aggregate compute above fits within a typical overnight processing window. This is a substantially narrower deployment envelope than the 143 ms/batch figure an earlier, unimplemented estimate of this design had assumed, and the gap matters: a design assumed to support near-real-time batch risk scoring in fact supports only workflows tolerant of multi-hour turnaround. The favorable 1,043 ms per-record figure additionally assumes full 512-record batches; a single-patient request still pays the full per-batch homomorphic evaluation cost, because CKKS cost depends on ciphertext count rather than on occupied slots, so per-record cost degrades by up to $512\times$ under sparse arrival. The same flat-cost property cuts the other way and we did not test it: at $n = 2^{15}$, 16,384 SIMD slots are available, $32\times$ more than the 512-record batches used throughout this paper’s measurements (chosen to divide our 4,228-record test set into round batches, not from a throughput analysis), and since per-ciphertext circuit latency does not depend on slot occupancy, batching closer to the full slot count could in principle reduce the reported per-record latency, and with it the $7.40 \times 10^6\times$ overhead this paper calls the binding deployment constraint, by up to another order of magnitude. We did not measure this because it would require rerunning the full encrypted pipeline, which Sec. 6 already establishes as costly at the batch size used here; we flag it as an unexplored, potentially consequential lever on this paper’s own headline result rather than assume 512 was the throughput-optimal

choice. Deployments must therefore aggregate requests into batching windows, accepting the induced delay. Compiler-optimized HE toolchains such as HEaAN.MLIR [71] may reduce the constant factor, and our measurement itself used only 8 of the host’s cores under heavy contention from other tenants on a shared research cluster (Sec. 4.4); dedicated, adequately parallel hardware would very likely reduce wall-clock latency substantially, though by how much remains to be measured rather than assumed. Dedicated hardware acceleration is a more aggressive alternative to additional CPU cores: Hu et al.’s ALT [72] reports an area-efficient, low-latency FPGA design for a different leveled scheme (TFHE) that reduces per-bootstrap latency by roughly an order of magnitude over CPU baselines in their evaluation, and Chen et al.’s FHE4DMM [73] specifically targets low-latency distributed matrix multiplication under FHE, the exact operation Sec. 6’s ablation identifies as this architecture’s dominant cost (the four linear layers, not the activations); neither result is directly portable to this paper’s CKKS/SEAL, CPU-only, single-node setting without further implementation work, so we report them as evidence that the latency this paper measures is not believed to be a hard floor for the field, not as a projection of what HE-MEDINFER itself would achieve on such hardware. We also do not evaluate whether an overnight batch is clinically actionable for the specific task benchmarked here: in-hospital mortality risk from the first 24 hours of an ICU stay is, in most clinical workflows, used for early triage and resourcing decisions made *during* that stay, not as a once-daily retrospective report. Whether a score computed hours after the window it describes still supports the decisions this benchmark task is meant to inform is a clinical-workflow question this paper does not answer, and it should be resolved before “overnight batch” is treated as a sufficient deployment envelope rather than merely a computationally feasible one. Separately, because meaningful throughput currently depends on dedicated, adequately parallel (or GPU) hardware beyond what this project had available, institutions without that infrastructure would face disproportionately worse latency under this design; we do not evaluate resource-constrained deployment settings, but flag the risk that HE-based compliance could become disproportionately available to better-resourced institutions.

Single-run measurements for the encrypted pipeline. Unlike the plaintext and DP-SGD baselines (10 independently trained and evaluated models each), HE-MEDINFER and the Lee et al. proxy are each reported from a single real run of the encrypted pipeline on the full test set, because the computational cost characterized above made replication infeasible within this project’s available compute. We report point estimates rather than distributions for these two methods and do not claim statistical significance for comparisons involving them (Sec. 6); this is a genuine, currently open limitation of the evidence in this paper, not a stylistic choice.

Evidence from synthetic data. As stated in Section 6.1, all diagnostic-utility results are computed on a generator fitted to published MIMIC-III statistics rather than on the records themselves. The systems and cryptographic results transfer unchanged; the AUROC values do not, and validation on credentialed MIMIC-III access is the immediate next step. Even with that access, MIMIC-III represents a single U.S. tertiary academic center, and generalization to community

hospitals, non-ICU settings, or non-English-language EHR systems would remain open.

Integrity under a malicious server. As analyzed in Section 3, confidentiality holds against an actively malicious Inference Server but correctness does not. A server that returns an arbitrary well-formed ciphertext produces a decryptable, clinically wrong prediction that the audit chain records faithfully and does not flag. We regard closing this gap through verifiable computation over ciphertexts as the most important open problem for clinical deployment of this class of system. Recent work on verifiable HE, such as Mahato and Chakraborty’s fast verifiable FHE technique for cloud-hosted secret computation [74], targets exactly this gap, though at a proof-generation overhead that has not, to our knowledge, been characterized for a multi-layer CKKS-MLP circuit at the depth this paper’s architecture requires; whether such techniques remain practical once stacked on top of the latency this paper already measures is an open question, not one this paper answers.

Subgroup performance. We report only pooled metrics. Because the synthetic cohort reproduces the demographic composition of MIMIC-III, which is documented to carry outcome disparities across race and insurance status, a pooled AUROC of 0.8622 may conceal materially worse performance in under-represented subgroups. We did not measure this, and we note a specific mechanism by which HE could plausibly worsen it: the Chebyshev approximation error is largest at the extremes of the pre-activation range, which are disproportionately occupied by atypical physiological presentations, so approximation-induced degradation may fall unevenly across subgroups. We are precise about what our own evidence could and could not have shown: the synthetic cohort is demographically parameterized, so a within-cohort comparison of the plaintext-vs-HE AUROC gap across its subgroup labels was, in principle, computable from data already generated for this study, and would have at least bounded whether the hypothesized mechanism is present in this specific pipeline, even though it is a synthetic cohort and so could not establish real clinical equity. We did not run that comparison, and we do not present it here rather than construct it post hoc without pre-registered subgroup definitions; we flag its omission as a concrete, immediately actionable gap rather than defer it entirely to future work requiring credentialed real-data access. Any clinical equity claim, however, still requires subgroup-stratified evaluation on real records.

7.4 Threats to Validity

Internal validity. Hyperparameter choices (degree-7 Chebyshev, $n = 2^{15}$, the only parameter set that supports degree-7 at 128-bit security, Sec. 4.4) were selected by grid search on a held-out 10% validation split carved from the 80% train partition (seed = 42, disjoint from the 20% test split reported in Table 4), so the reported test metrics are not tuned on. The residual threat is that the generator and the model were specified by the same authors, so a mis-specification shared between them would not be detected by internal comparison.

External validity. The dominant threat is the synthetic cohort (Section 6.1): diagnostic-utility results establish internal consistency, not clinical performance. Beyond that,

MIMIC-III represents a single U.S. tertiary academic center and results may differ for community hospitals or non-ICU settings.

Construct validity. HEPI is a proposed construct, not a validated one. Its weights are a stated design position rather than an elicited optimum; the sensitivity analysis in Section 6 shows that HE-MEDINFER and the Lee et al. proxy show no meaningful, tested separation across the entire weight simplex, so at minimum the metric does not manufacture a comparative ranking where the underlying single-run measurement supports none. More seriously, the ablation study demonstrates that HEPI is blind to auditability and, separately, to per-record batching efficiency (removing SIMD packing leaves the composite numerically unchanged despite a real $512\times$ per-record throughput cost, Sec. 6, because the overhead term is defined on per-circuit rather than per-record-delivered latency), so it should be read as summarizing the security-utility-overhead trade-off it explicitly encodes and not as a general measure of deployment readiness. A related, less visible limitation is that HEPI’s log-scaled overhead term is heavily compressive at the overhead magnitudes this paper measures: the framework’s own $7.40 \times 10^6\times$ latency overhead, which Sec. 7 calls the binding deployment constraint, costs only 0.025 of the term’s possible 0.200 (Sec. 7), so a catastrophic, deployment-disqualifying latency still yields a superficially high composite score. This is not merely a hypothetical failure mode: Fig. 1’s `Verify()` function takes τ_{HEPI} as a required, operator-supplied input and ships no numeric default, precisely because a shipped default is what makes a defect like this one silently exploitable; but if an operator naively supplies $\tau_{\text{HEPI}} = 0.70$ as one plausible illustrative example threshold, the measured $\text{HEPI} = 0.825$ clears that threshold, and the gate would issue a “Confirmed” deployment-readiness verdict for a system this paper’s own Discussion (Sec. 7) and Limitations (Sec. 7) call unsuitable for real-time or even same-hour clinical use. We view this as a genuine defect of the metric as currently weighted, not a result to be defended: HEPI should be read as a diagnostic decomposition of where a design’s cost falls, and no threshold value, illustrative or otherwise, should be relied upon to certify deployment readiness without an operator first reweighting β to reflect how much latency actually matters for their use case.

Conclusion validity. With 10 runs per arm and complete separation between the two distributions ($r = 1.00$), the Mann-Whitney result is not in question, though we note the reported power is computed from the observed effect rather than from an independent design target and should be read as a description of what was found, not a pre-registered power guarantee. The ablation and weight-sensitivity analyses are reported descriptively without significance testing, recombine the same single-run inputs used elsewhere in this paper, and should be read as such.

8 CONCLUSION AND FUTURE WORK

We presented HE-MEDINFER, a three-tier homomorphic encryption framework for privacy-preserving healthcare ML inference without inference-time data decryption, and implemented it end to end: real CKKS encryption and ho-

homomorphic inference via Microsoft SEAL, not a numerical simulation. We also introduced the HEPI metric for principled evaluation of HE-based clinical AI deployments. On a synthetic cohort calibrated to published MIMIC-III statistics for in-hospital mortality prediction, HE-MEDINFER attains an AUROC of 0.8622 under full CKKS encryption against 0.8607 in plaintext on the same trained model, a small, single-run gap that is nonetheless meaningfully smaller than an earlier, unimplemented estimate of this pipeline had assumed, with a measured composite HEPI of 0.825. HE-MEDINFER and the Lee et al. proxy baseline show no meaningful separation on this single-run measurement (0.8622 vs. 0.8623 AUROC, identical HEPI to four decimal places, no significance test performed; Sec. 6); HE-MEDINFER's contribution over that baseline is therefore architectural and regulatory, not a demonstrated accuracy advantage, and we report the corrected comparison rather than an earlier, unimplemented projection of a 0.033-AUROC margin. HE-MEDINFER provides a compliance mapping to the HIPAA technical safeguards, GDPR Articles 25 and 32, and NIST SP 800-175B [15] that the Lee et al. proxy does not, though this mapping does not address computational integrity under a malicious server (Sec. 5).

We are deliberate about what implementing this system, rather than only designing it, actually established. Accuracy under encryption is cheap in absolute terms though not demonstrably free given single-run evidence (Sec. 6); latency is not: real CKKS execution measured $7.40 \times 10^6 \times$ slower than plaintext on the same architecture, five to six orders of magnitude beyond what an unimplemented design estimate had assumed, and $n = 2^{13}$, one of the three parameter points the original design treated as interchangeable, turned out to be cryptographically infeasible for this architecture at any activation degree, a finding only implementation surfaces (Sec. 4.4). The diagnostic-utility results establish that the pipeline is internally consistent and reproducible, and quantify the AUROC lost to homomorphic evaluation on paired inputs, but they are not evidence of clinical performance on real patients, and the encrypted-pipeline results reported here are single runs rather than replicated distributions, for the same computational-cost reasons.

Future directions follow directly from the limitations identified above: (i) evaluation on credentialed MIMIC-III access with subgroup-stratified metrics, which is a precondition for any clinical claim; (ii) replicated independent runs of the encrypted pipeline on dedicated (non-shared) compute, to support the kind of significance testing this paper could establish for the plaintext and DP-SGD baselines but not for HE-MEDINFER itself; (iii) verifiable computation over ciphertexts to close the integrity gap under a malicious inference server, which we regard as the most consequential open problem; (iv) GPU-accelerated HE arithmetic and dedicated, uncontended parallel hardware to address the four-to-five-order-of-magnitude latency overhead measured in Sec. 6, which is a substantially larger target than the sparse-arrival penalty an unimplemented estimate had anticipated; (v) threshold HE supporting multi-institutional inference without a central key authority; (vi) extending HEPI to represent auditability and demographic parity, both of which the current formulation cannot see; and (vii) evaluation on imaging modalities using polynomial-approximated CNN

inference.

ACKNOWLEDGMENT

Data availability and use. This study did not access the MIMIC-III Clinical Database directly. All reported results were computed on synthetic records generated from *published, aggregate* summary statistics of MIMIC-III [16], [65], as described in Section 6.1; no individual-level patient data, restricted-access credential, or PhysioNet Data Use Agreement was required or obtained for the work reported here. Researchers seeking to reproduce this study on the underlying records must independently complete CITI training and execute the PhysioNet Credentialed Health Data Use Agreement with MIT Laboratory for Computational Physiology. The synthetic generator, fitted parameters, seeds, and evaluation harness are released under the MIT license and require no credentialing.

The authors acknowledge the use of Claude (Anthropic) as a language model assistant during the preparation of this manuscript. The AI tool was used exclusively to support literature synthesis, text revision, and structural organization. All scientific content, methodological decisions, data interpretation, and intellectual contributions remain the sole responsibility of the human authors. This disclosure follows the ethical guidelines of IEEE for AI-assisted academic writing.

Institutional acknowledgments withheld for double-blind review; will be inserted upon acceptance.

REFERENCES

- [1] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, "AI in health and medicine," *Nature Medicine*, vol. 28, pp. 31–38, 2022.
- [2] E. J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nature Medicine*, vol. 25, pp. 44–56, 2019.
- [3] A. Esteva, A. Robicquet, B. Ramsundar, V. Kuleshov, M. DePristo, K. Chou, C. Cui, G. Corrado, S. Thrun, and J. Dean, "A guide to deep learning in healthcare," *Nature Medicine*, vol. 25, pp. 24–29, 2019.
- [4] R. Miotto, F. Wang, S. Wang, X. Jiang, and J. T. Dudley, "Deep learning for healthcare: review, opportunities and challenges," *Briefings in Bioinformatics*, vol. 19, no. 6, pp. 1236–1246, 2018.
- [5] M. E. H. Chowdhury, T. Rahman, A. Khandakar, R. Mazhar, and et al., "Can AI help in screening viral and COVID-19 pneumonia?" *IEEE Access*, vol. 8, pp. 132 665–132 676, 2020.
- [6] Z. L. Teo, A. J. Thirunavukarasu, K. Elangovan, H. Cheng, P. Moova, and B. Soetikno, "Generative artificial intelligence in medicine," *Nature Medicine*, 2025.
- [7] W. N. Price and I. G. Cohen, "Privacy in the age of medical big data," *Nature Medicine*, vol. 25, pp. 37–43, 2019.
- [8] I. Y. Chen, P. Szolovits, and M. Ghassemi, "Can AI help reduce disparities in general medical and mental health care?" *AMA Journal of Ethics*, vol. 21, no. 2, pp. 167–179, 2019.
- [9] G. Kaissis, M. R. Makowski, D. Rueckert, and R. F. Braren, "End-to-end privacy preserving deep learning on multi-institutional medical imaging," *Nature Machine Intelligence*, vol. 3, pp. 473–484, 2021.
- [10] P. F. Edemekong, P. Annamaraju, and M. J. Haydel, "Health insurance portability and accountability act (HIPAA)," *StatPearls Publishing* (NCBI Bookshelf ID: NBK500019), 2023. [Online]. Available: <https://www.ncbi.nlm.nih.gov/books/NBK500019/>
- [11] P. Voigt and A. von dem Bussche, *The EU General Data Protection Regulation (GDPR): A Practical Guide*. Springer International Publishing, 2017.

- [12] C. Gentry, "Fully homomorphic encryption using ideal lattices," in *Proceedings of the Forty-First Annual ACM Symposium on Theory of Computing (STOC '09)*, 2009.
- [13] R. Gilad-Bachrach, N. Dowlin, K. Laine, K. Lauter, M. Naehrig, and J. Wernsing, "CryptoNets: Applying neural networks to encrypted data with high throughput and accuracy," in *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, PMLR vol. 48, 2016. [Online]. Available: <https://proceedings.mlr.press/v48/gilad-bachrach16.html>
- [14] J.-W. Lee, H. Kang, Y. Lee, W. Choi, J. Eom, M. Deryabin, E. Lee, J. Lee, D. Yoo, Y.-S. Kim, and J.-S. No, "Privacy-preserving machine learning with fully homomorphic encryption for deep neural network," *IEEE Access*, vol. 10, pp. 30 039–30 054, 2022.
- [15] National Institute of Standards and Technology, "Guideline for using cryptographic standards in the federal government: Cryptographic mechanisms," NIST Special Publication 800-175B Revision 1, Tech. Rep., 2020.
- [16] A. E. W. Johnson, T. J. Pollard, L. Shen, L.-w. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. A. Celi, and R. G. Mark, "MIMIC-III, a freely accessible critical care database," *Scientific Data*, vol. 3, p. 160035, 2016.
- [17] H. B. Mann and D. R. Whitney, "On a test of whether one of two random variables is stochastically larger than the other," *The Annals of Mathematical Statistics*, vol. 18, no. 1, pp. 50–60, 1947.
- [18] J. H. Cheon, A. Kim, M. Kim, and Y. Song, "Homomorphic encryption for arithmetic of approximate numbers," in *Advances in Cryptology – ASIACRYPT 2017*, 2017.
- [19] J. Fan and F. Vercauteren, "Somewhat practical fully homomorphic encryption," IACR Cryptology ePrint Archive, Report 2012/144, Tech. Rep., 2012. [Online]. Available: <https://eprint.iacr.org/2012/144>
- [20] C. Boura, N. Gama, M. Georgieva, and D. Jetchev, "CHIMERA: Combining ring-LWE-based fully homomorphic encryption schemes," *Journal of Mathematical Cryptology*, 2020.
- [21] A. Wood, K. Najarian, and D. Kahrobaei, "Homomorphic encryption for machine learning in medicine and bioinformatics," *ACM Computing Surveys*, vol. 53, no. 4, pp. 1–35, 2020.
- [22] Microsoft SEAL Team, "Microsoft SEAL (release 4.1)," 2024. [Online]. Available: <https://github.com/microsoft/SEAL>
- [23] P. Mohassel and Y. Zhang, "SecureML: A system for scalable privacy-preserving machine learning," in *IEEE Symposium on Security and Privacy (S&P)*, 2017.
- [24] F. Boemer, R. P. Cammarota, D. Demmler, T. Schneider, and H. Yalame, "MP2ML: A mixed-protocol machine learning framework for private inference," in *Proceedings of the 2020 Workshop on Privacy-Preserving Machine Learning in Practice (PPMLP)*, 2020.
- [25] W.-j. Lu, Z. Huang, C. Hong, Y. Ma, and H. Qu, "Pegasus: Bridging polynomial and non-polynomial evaluations in homomorphic encryption," *IEEE Symposium on Security and Privacy*, 2021.
- [26] T. Ryffel, A. Trask, M. Dahl, B. Wagner, J. Mancuso, D. Rueckert, and J. Passerat-Palmbach, "A generic framework for privacy preserving deep learning," in *Privacy in Machine Learning Workshop, NeurIPS*, 2018.
- [27] S. Wagh, S. Tople, F. Benhamouda, E. Kushilevitz, P. Mittal, and T. Rabin, "FALCON: Honest-majority maliciously secure framework for private deep learning," *Proceedings on Privacy Enhancing Technologies*, vol. 2021, no. 1, pp. 188–208, 2021.
- [28] O. Mudannayake, A. Indika, U. Jayasinghe, G. M. Lee, and J. Alawatugoda, "On privacy-preserved machine learning using secure multi-party computing: Techniques and trends," *Computers, Materials & Continua*, 2025.
- [29] J. Yuan, W. Liu, J. Shi, and Q. Li, "Approximate homomorphic encryption based privacy-preserving machine learning: A survey," *Artificial Intelligence Review*, vol. 58, p. 82, 2025.
- [30] W. J. Buchanan and H. Ali, "Evaluation of privacy-preserving support vector machine (SVM) learning using homomorphic encryption," *Cryptography*, vol. 9, no. 2, p. 33, 2025.
- [31] F. Hendaoui, L. Trabelsi, A. Ferchichi, R. Meddeb, and R. Ahmed, "EPAD: Empowering privacy-preserving anomaly detection through homomorphic encryption and machine learning," *Cluster Computing*, vol. 28, p. 683, 2025.
- [32] L. Liu, X. Du, M. Ma, and D. Wang, "A privacy protection strategy based on homomorphic encryption and neural networks," in *Proceedings of the 2024 8th International Conference on Electronic Information Technology and Computer Engineering (ICEITCE)*, 2024.
- [33] Y. Joo, S. Ha, H. Oh, and Y. Paek, "Efficient keyset design for neural networks using homomorphic encryption," *Sensors*, vol. 25, no. 14, p. 4320, 2025.
- [34] T. Zhu, D. Ye, W. Wang, W. Zhou, and P. S. Yu, "More than privacy: Applying differential privacy in key areas of artificial intelligence," *IEEE Transactions on Knowledge and Data Engineering*, vol. 34, no. 6, pp. 2824–2843, 2022.
- [35] S. Sav, A. Pyrgelis, J. R. Troncoso-Pastoriza, D. Froelicher, J.-P. Bossuat, J. S. Sousa, and J.-P. Hubaux, "POSEIDON: Privacy-preserving federated neural network learning," *Proceedings of NDSS*, 2021.
- [36] D. Froelicher, J. R. Troncoso-Pastoriza, A. Pyrgelis, S. Sav, J. S. Sousa, J.-P. Bossuat, and J.-P. Hubaux, "Scalable privacy-preserving distributed learning," in *Proceedings on Privacy Enhancing Technologies*, 2021.
- [37] J. Ma, S.-A. Naas, S. Sigg, and X. Lyu, "Privacy-preserving federated learning based on multi-key homomorphic encryption," *International Journal of Intelligent Systems*, vol. 37, no. 9, pp. 5880–5901, 2022.
- [38] J. Shen, Y. Zhao, S. Huang, and Y. Ren, "Secure and flexible privacy-preserving federated learning based on multi-key fully homomorphic encryption," *Electronics*, vol. 13, no. 22, p. 4478, 2024.
- [39] R. Xu, N. Baracaldo, Y. Zhou, A. Anwar, and H. Ludwig, "HybridAlpha: An efficient approach for privacy-preserving federated learning," *ACM Workshop on Artificial Intelligence and Security*, 2019.
- [40] Y. Chang, K. Zhang, J. Gong, and H. Qian, "Privacy-preserving federated learning via functional encryption, revisited," *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 1855–1869, 2023.
- [41] G. Xu, H. Li, S. Liu, K. Yang, and X. Lin, "Verifynet: Secure and verifiable federated learning," *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 911–926, 2020.
- [42] F. Haq, C. Chen, and Z. Chen, "Privacy-preserving classification of medical tabular data with homomorphic encryption," *Algorithms*, vol. 18, no. 12, p. 731, 2025.
- [43] V. S. Naresh and S. Reddi, "Exploring the future of privacy-preserving heart disease prediction: A fully homomorphic encryption-driven logistic regression approach," *Journal of Big Data*, vol. 12, p. 56, 2025.
- [44] T. Bhattacharjee, A. Haque, F. Shamim, and D. De, "FedHealthcare: Federated learning and lightweight additive homomorphic encryption-based privacy-preserving healthcare," *Security and Privacy*, vol. 8, no. 6, 2025.
- [45] M. Kim, Y. Song, S. Wang, Y. Xia, and X. Jiang, "Secure logistic regression based on homomorphic encryption: Design and evaluation," *JMIR Medical Informatics*, vol. 6, no. 2, p. e19, 2018.
- [46] A. Vizitiu, C. I. Nită, A. Puia, C. Suciu, and L. M. Itu, "Applying deep neural networks over homomorphic encrypted medical data," *Computational and Mathematical Methods in Medicine*, 2020.
- [47] O. Kocabas and T. Soyata, "Utilizing homomorphic encryption to implement secure and private medical cloud computing," in *2015 IEEE 8th International Conference on Cloud Computing (CLOUD)*, 2015.
- [48] J. L. Raisaro, J. Troncoso-Pastoriza, M. Misbach, J. S. Sousa, S. Pradervand, E. Missiaglia, O. Michielin, B. Ford, and J.-P. Hubaux, "MedCo: Enabling secure and privacy-preserving exploration of distributed clinical and genomic data," *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol. 16, no. 4, pp. 1328–1341, 2018.
- [49] M. Zuber and R. Sirdey, "Efficient homomorphic evaluation of k -NN classifiers," *Proceedings on Privacy Enhancing Technologies*, vol. 2021, no. 2, 2021.
- [50] J. W. Bos, K. Lauter, J. Loftus, and M. Naehrig, "Improved security for a ring-based fully homomorphic encryption scheme," in *Cryptography and Coding – IMACC 2013*, 2013, pp. 45–64.
- [51] M. R. Albrecht, R. Player, and S. Scott, "On the concrete hardness of learning with errors," *Journal of Mathematical Cryptology*, vol. 9, no. 3, pp. 169–203, 2015.
- [52] H. Goganaboina and H. Wu, "Parameter settings and efficient computation for homomorphic encryption in CKKS," *Electronics*, vol. 15, no. 11, p. 2391, 2026.
- [53] J. H. Cheon, K. Han, A. Kim, M. Kim, and Y. Song, "Bootstrapping for approximate homomorphic encryption," in *Advances in Cryptology – EUROCRYPT 2018*, 2018.
- [54] J. H. Ju, J. Park, J. Kim, M. Kang, D. Kim, and J. H. Cheon, "Neu-Jeans: Private neural network inference with joint optimization of

- convolution and FHE bootstrapping,” in *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2024.
- [55] A. Pirillo and L. Colombo, “ReBoot: Encrypted training of deep neural networks with CKKS bootstrapping,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 29, pp. 24 837–24 845, 2026.
- [56] S. Gao and K. Yates, “Leveled homomorphic encryption schemes for homomorphic encryption standard,” *Journal of Mathematical Cryptology*, vol. 19, no. 1, p. 20240024, 2025.
- [57] J. Wan, Y. Liao, Y. Liao, Z. Tan, J. Xu, Z. L. Jiang, and B. Fang, “CipherAPR: Accelerating RNS-CKKS encrypted inference via importance-guided and level-aware mixed-degree polynomial design,” *Cryptography*, vol. 10, no. 4, p. 44, 2026.
- [58] Y. Xie, R. Huang, and J. Qiu, “Verifiable threshold multi-party fully homomorphic encryption from share resharing,” *Applied Sciences*, vol. 15, no. 9, p. 4745, 2025.
- [59] X. Wan, H. Lin, M. Wang, and W. Shen, “Hierarchical threshold multi-key fully homomorphic encryption,” *Journal of Information Security and Applications*, vol. 89, p. 103919, 2025.
- [60] C. Peikert, “A decade of lattice cryptography,” *Foundations and Trends in Theoretical Computer Science*, vol. 10, no. 4, pp. 283–424, 2016.
- [61] National Institute of Standards and Technology, “Module-lattice-based key-encapsulation mechanism standard (FIPS 203),” *NIST Federal Information Processing Standards*, 2024.
- [62] J. Bos, C. Costello, L. Ducas, I. Mironov, M. Naehrig, V. Nikolaenko, A. Raghunathan, and D. Stebila, “FRODO: Take off the ring! practical, quantum-secure key exchange from LWE,” *ACM CCS*, 2016.
- [63] B. Li and D. Micciancio, “On the security of homomorphic encryption on approximate numbers,” in *Advances in Cryptology – EUROCRYPT 2021*, 2021.
- [64] C. Riou, M. El Azzouzi, A. Hespel, E. Guillou, G. Coatrieux, and M. Cuggia, “Ensuring general data protection regulation compliance and security in a clinical data warehouse from a university hospital: Implementation study,” *JMIR Medical Informatics*, vol. 13, p. e63754, 2025.
- [65] H. Harutyunyan, H. Khachatrian, D. C. Kale, G. Ver Steeg, and A. Galstyan, “Multitask learning and benchmarking with clinical time series data,” *Scientific Data*, vol. 6, p. 96, 2019.
- [66] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, PMLR vol. 70, 2017, pp. 1321–1330. [Online]. Available: <https://proceedings.mlr.press/v70/guo17a.html>
- [67] S. van Buuren and K. Groothuis-Oudshoorn, “mice: Multivariate imputation by chained equations in R,” *Journal of Statistical Software*, vol. 45, no. 3, pp. 1–67, 2011.
- [68] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, “Deep learning with differential privacy,” in *ACM CCS*, 2016.
- [69] C. Dwork, F. McSherry, K. Nissim, and A. Smith, “Calibrating noise to sensitivity in private data analysis,” in *Theory of Cryptography Conference*, 2006.
- [70] H.-C. Wang and J.-L. Wu, “BlockFormer: An efficient private batch transformer inference with homomorphic encryption via block packing,” *IEEE Access*, vol. 14, pp. 74 048–74 066, 2026.
- [71] S. Park, W. Song, S. Nam, H. Kim, J. Shin, and J. Lee, “HEaAN.MLIR: An optimizing compiler for fast ring-based homomorphic encryption,” *Proceedings of the ACM on Programming Languages*, vol. 7, no. PLDI, 2023.
- [72] X. Hu, Z. Li, Z. Wang, and X. Lu, “ALT: Area-efficient and low-latency FPGA design for torus fully homomorphic encryption,” *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 32, no. 4, pp. 645–657, 2024.
- [73] Y. Chen, Q.-S. Hua, Z. Hong, L. Zhu, and H. Jin, “FHE4DMM: A low-latency distributed matrix multiplication with fully homomorphic encryption,” *IEEE Transactions on Parallel and Distributed Systems*, vol. 36, no. 4, pp. 645–658, 2025.
- [74] G. K. Mahato and S. K. Chakraborty, “A fast verifiable fully homomorphic encryption technique for secret computation on cloud data,” *International Journal of Information Technology*, vol. 17, no. 4, pp. 2193–2207, 2025.